Projector-Resolved Seed Calculus IX

Arithmetic Packets, Dyadic Towers, and Local Polynomial Deformation

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Abstract

Arithmetic packet coordinates connect congruences among Euler coefficients with reconstruction at finite and p-adic precision. We develop this connection, prove the stated Nishibuchi congruence conjecture in its full parameter range, and extend the framework to local polynomial deformation. The arithmetic part distinguishes normalized and unnormalized Euler packets, proves amplified difference bounds, gives exact dyadic displacements and least packet periods for four residue addresses, and records an optimal classical dyadic reconstruction tower. The lifting part identifies finite obstruction classes, shared corrections and correction fibers, constructs compatible polynomial towers, and proves quantitative persistence under simultaneous coefficient and parameter deformation. An analytic implicit function theorem supplies local solution graphs and computes their strict derivatives from the polynomial Jacobian. Every continuous local root selection through the same nonsingular point agrees with this graph near the base parameter. Source references identify the corresponding Lean proofs; classical derivations without a dedicated formal bridge are marked explicitly. General singular persistence, unrestricted presentation invariance, global descent and further exact valuation questions remain open within this program.

1 Scope and conventions

The central objects are sequences and polynomial equations equipped with arithmetic coordinates. Residue projectors select addresses in a sequence, finite differences measure compatibility between nearby addresses, and valuation bounds control the passage from finite precision to a p-adic limit. These constructions provide a common language for the Euler packet results and the later lifting results. Each theorem nevertheless retains its own coefficient ring, topology and nonsingularity hypotheses.

This edition integrates the original arithmetic framework with the completed extensions. The stated external Nishibuchi conjecture is proved in its full specified parameter range. The dyadic refinements and polynomial deformation theorems give stronger conclusions under the hypotheses stated with each result. The native conjectures retain their original identifiers, 13.2 and 13.3, when discussed as research questions, although the section numbering of this consolidated manuscript is new.

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. When p is prime, $\mathbb{Z}_p$ denotes the p-adic integers and $\mathbb{Q}_p$ their fraction field. We normalize $v_p(p) = 1$. For a rational or p-adic quantity, a displayed lower valuation bound is understood with the mathematical convention $v_p(0) = +\infty$. The corresponding formal predicates are zero-safe: the quantity is zero or its valuation meets the stated bound. An exact finite valuation assertion requires a nonzero quantity, and this is

stated where used. Congruence in $\mathbb{Z}_p$ means ideal divisibility. Thus

$$x \equiv y \pmod{p^r} \quad \Longleftrightarrow \quad x - y \in p^r \mathbb{Z}_p.$$

For vectors, ideal membership and congruence are coordinatewise.

Write $\Delta_u f(n) = f(n+u) - f(n)$ and use Δ_u^k for its k-fold iterate. Euler coefficients indexed by ordinary degree and by packet row are distinguished explicitly. A packet displacement by u rows is a degree displacement by Nu in a packet of width N.

The main text presents theorem statements and their arguments; accompanying source maps identify the Lean declarations and archived files that support them. The verification record keeps the original dates and scope of each build. In particular, the latest 91-check polynomial build covers the polynomial development, while earlier builds support the arithmetic results. This edition revises the exposition without adding Lean theorems or making new claims of historical priority. The analytic inverse-function results used in Section 5 come from Mathlib.

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2 Arithmetic packets and coefficient transport

We work with sequences, formal power series, and finite recurrence states. Residue selectors isolate coefficients in a chosen congruence class, while triangular transport expresses the same object in new coordinates. A valuation filtration determines how many coordinates are needed at a given precision. Together, these operations provide a common framework for the Euler congruences and the finite lifting constructions developed later.

The original arithmetic seed notation

$$\mathcal{S} = (A, a_0, \mathcal{O}, \mathcal{R}, \{P_{N,j}\}, w, \tau_p)$$

records a coefficient ring or module, a seed, allowed operators, relations, residue selectors, a depth weight, and a topology. This notation describes the ingredients of the construction; a category of seed presentations and a notion of equivalence still require separate definitions. We therefore state each theorem for a concrete object. Section 6 discusses the additional structure needed for the original native conjectures.

The Lean declarations cited in this section have namespace prefix `KomorebiPaperIX`. Their exact modules and source hashes are recorded in the accompanying statement map. All belong to the preserved baseline formalization; this section brings their statements and proofs together.

2.1 Residue selectors and exact reconstruction

Let R be a commutative ring, $a : \mathbb{N} \to R$, $N \geq 1$, and $0 \leq j < N$. Define

$$(P_{N,j}a)(n) = \begin{cases} a(n), & n \equiv j \pmod{N}, \\ 0, & \text{otherwise.} \end{cases}$$

This definition keeps the original index set. The compressed subsequence $a(Nq + j)$, used below for recurrence packets, is a different representation of the selected coefficients.

Theorem 2.1 (Residue reconstruction)**.** *The selectors satisfy*

$$P_{N,j}^2 = P_{N,j}, \qquad P_{N,i}P_{N,j} = 0 \quad (i \neq j), \qquad \sum_{j=0}^{N-1} P_{N,j} = I.$$

If $N \mid M$ and $M > 0$, then

$$P_{N,j}a = \sum_{\substack{0 \leq r < M \\ r \equiv j \pmod{N}}} P_{M,r}a.$$

The same identities hold coefficientwise for formal power series.

Proof. At each index n, exactly one standard residue label equals $n \bmod N$. Consequently a repeated selector changes nothing, two different selectors have zero composite, and the sum retains $a(n)$ exactly once. If $N \mid M$, reducing $n \bmod M$ modulo N gives $n \bmod N$. This proves the refinement identity at every coefficient. Equality of all coefficients proves the formal-series versions. □

Formal correspondence. `ResiduePackets.packet_idempotent`, `ResiduePackets.packet_orthogonal`, `ResiduePackets.packet_reconstruction`, and `ResiduePackets.packet_refinement`; the series variants have the prefix `series_packet` in the same module. The algebraic statements need no convergence hypothesis.

2.2 Formal exponential coordinates and Hurwitz multiplication

Over a characteristic-zero field K, put

$$\mathcal{E}(a) = \sum_{n\geq 0} \frac{a_n}{n!} t^n, \qquad \mathcal{D}(A)_n = n![t^n]A.$$

These inverse maps between sequences and $K[[t]]$ provide *formal* exponential coordinates. No analytic convergence is needed for the coefficient identities below.

Theorem 2.2 (Constructed unit-triangular coordinates)**.** *Let $h : \mathbb{N} \to K$ satisfy $h_0 = 1$, write $H = \mathcal{E}(h)$ and $H^{-1} = \mathcal{E}(k)$, and define*

$$U_H a = \mathcal{D}(H\mathcal{E}(a)).$$

Then U_H is an equivalence, with inverse $U_H^{-1} a = \mathcal{D}(H^{-1}\mathcal{E}(a))$. Its coefficient formulas are

$$(U_H a)_n = \sum_{r=0}^{n} \binom{n}{r} h_r a_{n-r}, \qquad (U_H^{-1} a)_n = \sum_{r=0}^{n} \binom{n}{r} k_r a_{n-r}.$$

The resulting triangular matrix has diagonal entries 1. The transported selectors

$$\widehat{P}_{N,j} = U_H P_{N,j} U_H^{-1}$$

are idempotent and pairwise orthogonal, and their sum is the identity.

Proof. The factorials are nonzero in K, so encoding and decoding cancel coefficientwise. The constant coefficient of H is 1; hence its formal power-series inverse exists. Multiplication by H and H^{-1} gives the two inverse laws. The ordinary product coefficient formula, multiplied by $n!$, gives the displayed binomial sums. In row n, the coefficient of a_n is $h_0 = 1$ and coefficients of a_q with $q > n$ vanish. Conjugating the identities of Theorem 2.1 by the constructed equivalence proves the transported projector identities. □

Formal correspondence. `EGF.decode_encode`, `EGF.encode_decode`, `EGF.egf_product`, `EGF.coordinateEquiv`, `EGF.coordinate_forward`, `EGF.coordinate_inverse`, and `EGF.transported_reconstruction`, together with the idempotence and orthogonality declarations in `EGFCoordinates`.

In characteristic p, factorial division is unavailable in general, but the coefficient operation remains meaningful over every commutative ring:

$$(f \star g)_n = \sum_{r=0}^{n} \binom{n}{r} f_r g_{n-r}, \qquad (D^d f)_n = f_{n+d}, \qquad \mathbf{1}_n = \begin{cases} 1, & n = 0, \\ 0, & n > 0. \end{cases}$$

Here $\star$ is Hurwitz multiplication. The symbol D^d denotes a coefficient shift, which agrees with formal differentiation after characteristic-zero encoding.

Theorem 2.3 (Hurwitz product rules)**.** *For $d \geq 0$,*

$$D^d(f \star g) = \sum_{r=0}^{d} \binom{d}{r} (D^r f) \star (D^{d-r} g).$$

If R has prime characteristic p, then

$$D^p(f \star g) = f \star D^p g + D^p f \star g.$$

Over a field, if $g_0 = 1$, a sequence b is constructed recursively with $b \star g = \mathbf{1}$. If also $D^p g = Dg$ in characteristic p, then $D^p b = Db$.

Proof. Pascal's identity proves the first product rule, and induction proves the higher rule. The intermediate binomial coefficients $\binom{p}{r}$ vanish in characteristic p. For the inverse, solve row n using the coefficient of b_n, which is $g_0 = 1$:

$$b_n = \mathbf{1}_n - \sum_{r<n} \binom{n}{r} b_r g_{n-r}.$$

Subtract the first and pth product rules for $b \star g = \mathbf{1}$. The assumed identity for g leaves $(D^p b - Db) \star g = 0$. A second triangular induction shows that convolution with a sequence having constant coefficient 1 has trivial kernel. □

Formal correspondence. `Hurwitz.shift_convolution`, `Hurwitz.prime_shift_convolution`, `Hurwitz.convolution_left_eq_zero`, `HurwitzInverse.inverse_convolution`, and `HurwitzInverse.inverse_prime_difference`. Working directly with coefficient sequences avoids the undefined expression $1/n!$ in $\mathbb{F}_p$ when $p \mid n!$.

2.3 Triangular inversion and a finite precision filtration

Theorem 2.4 (Forward substitution and denominator clearing)**.** *Let T_{nq} be coefficients over a field with $T_{nn} \neq 0$ for every n. For any sequence b, define*

$$a_n = \frac{b_n - \sum_{q<n} T_{nq} a_q}{T_{nn}}.$$

This constructs the unique solution of $\sum_{q=0}^{n} T_{nq} a_q = b_n$ for all n. If T_{nq} and b_n are integers and the solution is taken over $\mathbb{Q}$, then

$$\left(\prod_{r=0}^{n} T_{rr} \right) a_n \in \mathbb{Z}.$$

Proof. The recursion uses only smaller indices, and substitution verifies each row. If a second solution agrees through row $n-1$, subtracting row n and cancelling T_{nn} proves equality at n. For integrality, use induction and promote every earlier clearing product to $\prod_{r<n} T_{rr}$. Multiplication by the additional factor T_{nn} cancels the only new division. □

Formal correspondence. `Triangular.solve_row`, `Triangular.solve_unique`, and `Triangular.solve_denominator_integral`. The clearing product bounds the denominator; the unscaled solution can still be nonintegral.

To isolate the filtration argument, let p be any element of a commutative ring and let $w : \mathbb{N} \to \mathbb{N} \cup \{\infty\}$. Set $p^{[r]} = p^r$ for finite r and $p^{[\infty]} = 0$. The latter notation means the coefficient vanishes.

Theorem 2.5 (Filtered finite reconstruction)**.** *For every $n, L \in \mathbb{N}$ and sequences c, S in that ring,*

$$\begin{aligned} &\sum_{r=0}^{n} \binom{n}{r} c_r p^{[w(r)]} S_{n-r} \\ &\qquad \equiv \sum_{\substack{0 \le r \le n \\ w(r) < L}} \binom{n}{r} c_r p^{[w(r)]} S_{n-r} \pmod{p^L}. \end{aligned}$$

Here congruence means divisibility of the difference by p^L in the coefficient ring. In $\mathbb{Z}_p$, it means membership in $p^L \mathbb{Z}_p$.

Proof. The difference of the two sums consists exactly of terms with $w(r) \geq L$. Every such term is divisible by p^L, including the zero terms with infinite weight. Their finite sum has the same divisibility. □

Formal correspondence. `Filtered.filtered_triangular_reconstruction` and `Filtered.padic_filtered_reconstruction`. For weights $2r$, retaining $r \leq M$ gives the bound p^{2M+2} through `Filtered.even_weight_tail`. A sharper kernel weight can improve this bound, as in the classical dyadic reconstruction later in the paper. Each coefficient row is finite, so no hypothesis $w(r) \to \infty$ is needed here.

2.4 Finite Mahler coordinates and weighted differences

For $f : \mathbb{N} \to R$, define the step-h forward difference and its iterates by

$$\Delta_h f(n) = f(n+h) - f(n), \qquad \Delta_h^0 f = f,$$

and put $c_r(f) = \Delta_1^r f(0)$. Thus the coefficients are determined by the differences of f.

Theorem 2.6 (Finite Newton–Mahler identities)**.** *For every $n, s, h \in \mathbb{N}$,*

$$\begin{aligned} f(n) &= \sum_{r=0}^{n} c_r(f) \binom{n}{r}, \\ c_r(\Delta_1^s f) &= c_{s+r}(f), \\ \Delta_1^s f(n) &= \sum_{j=0}^{n} c_{s+j}(f) \binom{n}{j}, \\ \Delta_h^s f(n) &= \sum_{j=0}^{s} (-1)^{s-j} \binom{s}{j} f(n+jh). \end{aligned}$$

The finite interpolant $I_N f(n) = \sum_{r=0}^{N} c_r(f) \binom{n}{r}$ agrees with f for $n \leq N$. It has the same order-s unit-step difference at n whenever $n + s \leq N$. For arbitrary coefficients $c_0, \ldots, c_N$, the corresponding binomial interpolant has exactly those coefficients and zero coefficients above N.

Proof. Write $E_h f(n) = f(n+h)$ for the shift operator. The unit shift is $I + \Delta_1$. Expanding its nth power and evaluating at zero gives reconstruction. Composing difference operators gives the coefficient shift and applying reconstruction to $\Delta_1^s f$ gives the third identity. The binomial expansion of $(E_h - I)^s$ gives the explicit stencil. Binomial coefficients vanish when their lower index exceeds n, which proves the interpolation claim. Equality of every point of the stencil proves equality of differences. Repeated finite differences of $\binom{n}{r}$ at zero give the exact coefficient extraction. □

Formal correspondence. `Mahler.finite_mahler_reconstruction`, `Mahler.coefficient_difference`, `Mahler.difference_from_coefficients`, `Mahler.difference_explicit`, `Mahler.finite_sample_difference`, and `Mahler.coefficient_interpolant`. Finite samples determine the stencils contained in the sample interval; a universal tail estimate requires additional information.

For $x \in \mathbb{Z}_p$, extend the valuation to zero by setting

$$\operatorname{ord}_p(x) = \begin{cases} v_p(x), & x \neq 0, \\ \infty, & x = 0. \end{cases}$$

Thus $L \leq \operatorname{ord}_p(x)$ is equivalent to $x \in p^L \mathbb{Z}_p$.

Theorem 2.7 (Weighted difference bound)**.** *Let $f : \mathbb{N} \to \mathbb{Z}_p$ and let $w : \mathbb{N} \to \mathbb{N} \cup \{\infty\}$ satisfy $w(s+j) \le \mathrm{ord}_p(c_{s+j}(f))$ for $0 \le j \le n$. Then*

$$\mathrm{ord}_p(\Delta_1^s f(n)) \ge \min_{0\le j\le n} \left\{ w(s+j) + v_p\binom{n}{j} \right\}.$$

Proof. Apply the nonarchimedean inequality to the finite coefficient formula in Theorem 2.6. For $0 \le j \le n$, the integer $\binom{n}{j}$ is positive. The order of its product with $c_{s+j}(f)$ is the sum of the two orders, including the case where that coefficient is zero. □

Formal correspondence. `PadicBounds.nat_le_order_iff`, `PadicBounds.weighted_difference_bound`, and `PadicBounds.weighted_difference_divisibility`. An expansion on all of $\mathbb{Z}_p$ requires a convergence theorem in addition to these finite algebraic identities and bounds.

2.5 Unit modes and the dyadic distinction

Theorem 2.8 (Differences of unit modes)**.** *Over every commutative ring,*

$$\Delta_h^s(u^n) = u^n(u^h - 1)^s.$$

For any prime p, units $u_i \in \mathbb{Z}_p^\times$, coefficients $a_i \in \mathbb{Z}_p$, and a finite index set I, the sequence $f(n) = \sum_{i\in I} a_i u_i^n$ satisfies

$$\Delta_{p-1}^s f(n) \in p^s\mathbb{Z}_p \qquad (n, s \in \mathbb{N}).$$

Proof. Induction gives the geometric identity. Reduction of a unit to $\mathbb{F}_p^\times$ gives $u_i^{p-1} = 1$ modulo p. Taking s powers supplies p^s in each summand. The difference operator is linear over finite sums. □

Formal correspondence. `UnitModes.geometric_difference`, `UnitModes.unit_fermat_divisibility`, and `UnitModes.finite_unit_modes_divisibility`. The result includes $p = 2$, where the displayed step is 1. Its scope here is finite sums of modes.

Theorem 2.9 (Finite unit coordinates)**.** *Let p be prime, $L \ge 0$, and $a \in (\mathbb{Z}/p^{L+1}\mathbb{Z})^\times$. Define*

$$\omega_L(a) = a^{p^L}, \qquad \langle a\rangle_L = a\omega_L(a)^{-1}.$$

Then

$$a = \omega_L(a)\langle a\rangle_L, \qquad \omega_L(a)^{p-1} = 1, \qquad \langle a\rangle_L \equiv 1 \pmod p.$$

The two factors are uniquely determined by these properties and are compatible with reduction to lower precision. For $a \in (\mathbb{Z}/2^{L+2}\mathbb{Z})^\times$, there is instead a unique decomposition

$$a = \epsilon(a)\langle a\rangle_4, \qquad \epsilon(a) \in \{1, -1\}, \qquad \langle a\rangle_4 \equiv 1 \pmod 4.$$

Proof. The unit-group exponent $p^L(p-1)$ gives the root identity for ω_L. Its residue is that of a, so the remaining factor has residue 1. If two $(p-1)$st roots have equal residue, raising them to p^L makes them equal modulo p^{L+1}, while each root is fixed by this exponent. This proves uniqueness and then compatibility under reduction. In the dyadic construction, an odd residue is exactly 1 or -1 modulo 4. Choose that sign and multiply a by it to obtain the principal factor. The two signs remain distinct modulo 4, which proves uniqueness. □

Formal correspondence. `FiniteUnits.finite_unit_decomposition`, `FiniteUnits.decomposition_unique`, `UnitRefinement.omega_compatible`, `UnitRefinement.principal_compatible`, and `DyadicUnits.dyadic_unit_decomposition` with `DyadicUnits.decomposition_unique`. The finite $p = 2$ version of the first construction has trivial root-of-unity factor. The separate modulo-4 construction retains the sign that reduction modulo 2 cannot detect.

2.6 Recurrence packets and finite arithmetic synchronization

Theorem 2.10 (Companion jumps and packet recurrences)**.** *Let C be a $d \times d$ matrix over a commutative ring and $x_n = C^n x_0$. For $N, j, k \in \mathbb{N}$, put $b_k = x_{Nk+j}$. Then*

$$b_k = C^j (C^N)^k x_0, \qquad b_{k+1} = C^N b_k.$$

If $\chi_{C^N}(X) = X^d + \sum_{r<d} q_r X^r$, then

$$b_{k+d} = -\sum_{r=0}^{d-1} q_r b_{k+r}.$$

Every coordinate and every linear scalar observable satisfies the same recurrence. No invertibility or diagonalization hypothesis is required.

Proof. The jump identity follows from the power law for one matrix. Apply the Cayley–Hamilton identity $\chi_{C^N}(C^N) = 0$ to the state b_k. The leading coefficient is 1, so the final state can be isolated. A linear observable preserves the resulting finite sum. □

Formal correspondence. `Companion.companion_jump`, `Companion.packet_charpoly_recurrence`, and `Companion.packet_observable_recurrence`. The source-sequence versions derive the packet recurrence from the relation $x_{n+1} = Cx_n$. For the Fibonacci matrix $C = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, its jump trace is the Lucas number L_N and its determinant is $(-1)^N$. Thus

$$F_{N(k+2)+j} = L_N F_{N(k+1)+j} - (-1)^N F_{Nk+j}.$$

This specialization is `FibonacciPackets.fibonacci_packet_recurrence`; it is derived from the Fibonacci state.

Theorem 2.11 (Dyadic–odd packet factorization)**.** *Let $a \geq 0$ and let M be a positive odd integer. The Chinese remainder map is a ring isomorphism*

$$\mathbb{Z}/(2^a M)\mathbb{Z} \simeq \mathbb{Z}/2^a\mathbb{Z} \times \mathbb{Z}/M\mathbb{Z}.$$

For $0 \leq r < 2^a M$ and any coefficient sequence f,

$$P_{2^a M, r} f = P_{2^a, r \bmod 2^a} P_{M, r \bmod M} f.$$

The same identity holds for formal-series coefficient selectors.

Proof. The two moduli are coprime. An index has the prescribed combined residue exactly when it has both component residues. Applying this equivalence to the indicator defining each selector proves the formula. □

Formal correspondence. `CRTPackets.addressEquiv`, `CRTPackets.residue_equivalence`, `CRTPackets.packet_factorization`, and `CRTPackets.series_packet_factorization`. This theorem synchronizes finite arithmetic addresses. The passage from a family of p-adic objects to a global integer object is a separate question, discussed in Section 6.3.

2.7 Polyphase notation retained from the recovered manuscript

The recovered Annals text develops a broader polyphase vocabulary. We explain it through elementary identities with explicit algebraic derivations. These identities have not been separately formalized in Lean for this consolidation. For a sequence a and $M \geq 1$, let

$a_r^{[M]}(q) = a_{Mq+r}$ for $0 \le r < M$. If $A(z) = \sum_n a_n z^n$ and $A_r(w) = \sum_q a_{Mq+r} w^q$ are ordinary generating functions, then unique quotient and remainder give

$$A(z) = \sum_{r=0}^{M-1} z^r A_r(z^M), \qquad (a_r^{[M]})_s^{[N]} = a_{r+Ms}^{[MN]}.$$

For ordinary Cauchy multiplication, the phase-r product has a term without a carry and a term with one carry:

$$C_r(w) = \sum_{u+v=r} A_u(w)B_v(w) + w \sum_{u+v=r+M} A_u(w)B_v(w), \qquad 0 \le u, v < M.$$

Adding the two phase addresses gives the formula. It describes ordinary Cauchy multiplication; Hurwitz convolution has its own binomial weights.

For $M \ge 1$ and $s, r, k \in \mathbb{N}$, define the affine Mahler kernel by

$$\Theta_{M,s}(r,k) = [z^r](1+z)^s\big((1+z)^M - 1\big)^k = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \binom{Mj+s}{r}.$$

It vanishes unless $k \le r \le Mk + s$. For a finite binomial expansion $f(n) = \sum_{r=0}^{D} c_r \binom{n}{r}$, the kth Newton coefficient of $g(q) = f(Mq + s)$ is therefore

$$c_k(g) = \sum_{r=k}^{\min(D, Mk+s)} c_r \Theta_{M,s}(r,k).$$

Indeed, insert the finite stencil for $\Delta_1^k g(0)$ and interchange its two finite sums. Coefficient extraction of the binomial expansion of $((1+z)^M - 1)^k$ gives the kernel. In the binary zero-address case,

$$\Theta_{2,0}(k+j,k) = 2^{k-j} \binom{k}{j} \qquad (0 \le j \le k),$$

since $(2z + z^2)^k = z^k(2+z)^k$. Exact valuations for the general kernel remain to be determined. The recovered infinite analytic extensions and stronger historical identities lie beyond these finite derivations; their status is discussed in Section 6.5.

3 Congruential Euler packets and their arithmetic

Two related constructions run through this section. The congruential Euler family is indexed by a modulus N and an address j. Its exceptional dyadic member has $N = 4$ and four addresses. The classical signed Euler numbers arise from $N = 2$, $j = 0$. Keeping these indices visible distinguishes the four-packet period theorems from the classical edge truncations.

3.1 Original coefficients, normalized packets, and precision

Definition 3.1 (Congruential Euler coefficients)**.** For $N \ge 1$ and $0 \le j < N$, define rational coefficients by the formal identity

$$\sum_{d \ge 0} E_d^{(N,j)} \frac{t^d}{d!} = \left(\sum_{n \ge 0} \frac{t^{Nn}}{(Nn+j)!} \right)^{-1}. \tag{3.1}$$

The reciprocal exists because its denominator has constant coefficient $1/j! \neq 0$. Set

$$U_{N,j}(n) = E_{Nn}^{(N,j)}, \qquad C_{N,j}(n) = \frac{U_{N,j}(n)}{j!}.$$

Thus $C_{N,j}(0) = 1$ and $U_{N,j}(0) = j!$. All coefficients outside the multiples of N vanish.

The normalized source and its inverse are

$$K_{N,j}(t) = j! \sum_{n\geq 0} \frac{t^{Nn}}{(Nn+j)!}, \qquad \mathcal{P}_{N,j}(t) = \sum_{n\geq 0} C_{N,j}(n) \frac{t^{Nn}}{(Nn)!}. \tag{3.2}$$

Their product is one. Clearing factorials in its coefficient of t^{Nn} gives the triangular system

$$\sum_{q=0}^{n} \binom{Nn+j}{Nq} C_{N,j}(q) = \delta_{n0}. \tag{3.3}$$

In particular, for $n > 0$,

$$C_{N,j}(n) = -\binom{Nn+j}{Nn}^{-1} \sum_{q=0}^{n-1} \binom{Nn+j}{Nq} C_{N,j}(q).$$

The diagonal binomial coefficient is positive, so this recurrence determines the sequence uniquely. The Lean construction begins with this recurrence and derives the reciprocal identity from it. The resulting sequence is then identified with the degree-indexed coefficients of (3.1), so both descriptions refer to the same family.

The relevant declarations are `KomorebiPaperIX.Congruential.triangular_inverse`, `KomorebiPaperIX.Congruential.packet_is_actual_inverse`, and `KomorebiPaperIX.NishibuchiCompletion.generalizedEuler_at_packet`. The cleared identity with the original positive-address exponential section is `KomorebiPaperIX.NishibuchiCompletion.original_generating_identity`.

Proposition 3.2 (A denominator bound for every address)**.** *For $N \geq 1$, $0 \leq j < N$, and $n \in \mathbb{N}$, set*

$$D_{N,j}(n) = \prod_{r=1}^{n} \binom{Nr+j}{Nr}, \qquad D_{N,j}(0) = 1.$$

Then $D_{N,j}(n) C_{N,j}(n) \in \mathbb{Z}$.

Proof. Proceed by induction in the recurrence (3.3). For every $q < n$, the product $D_{N,j}(q)$ divides $D_{N,j}(n-1)$. Multiplying the recurrence by $D_{N,j}(n-1)$ therefore makes every previous term integral. The diagonal factor completes the product $D_{N,j}(n)$, proving the claim. At $n = 0$, both factors are one. This gives a uniform denominator bound without requiring integrality of the normalized coefficients themselves. The Lean statements are `KomorebiPaperIX.Congruential.clearing_product` and `KomorebiPaperIX.Congruential.universal_denominator_clearing`. □

Example 3.3 (The Bernoulli boundary address)**.** At $N = 2$, the address $j = 1$ gives a second classical family. Define the Bernoulli numbers by

$$B(z) = \frac{z}{e^z - 1} = \sum_{r\geq 0} B_r \frac{z^r}{r!}, \qquad B_0 = 1, \quad B_1 = -\frac{1}{2}.$$

All quotients here are regular formal series. The identity

$$2B(z) - B(2z) = \frac{z}{\sinh z} = \sum_{n\geq 0} C_{2,1}(n) \frac{z^{2n}}{(2n)!}$$

follows by putting the two fractions on a common denominator and using (3.1). Comparing even coefficients yields

$$C_{2,1}(n) = (2 - 2^{2n})B_{2n} \qquad (n \in \mathbb{N}).$$

This classical derivation restores the boundary example from the original manuscript. No dedicated Lean bridge for this Bernoulli specialization was identified in the preserved sources.

Definition 3.4 (Differences and valuation conventions)**.** For a sequence f and $h, k, n \in \mathbb{N}$, put

$$\Delta_h f(n) = f(n+h) - f(n), \qquad \Delta_h^0 f(n) = f(n), \qquad \Delta_h^{k+1} f(n) = \Delta_h^k f(n+h) - \Delta_h^k f(n).$$

We use the extended valuation $v_p(0) = +\infty$ in mathematical statements. Thus $v_p(x) \geq L$, including negative integers L, means $x = 0$ or the ordinary valuation of the nonzero rational x is at least L. For integral coefficients and $L \geq 0$, it is equivalent to $x \in p^L\mathbb{Z}_p$. In expressions involving a step multiplier we write $\nu_p(u) = v_p(u)$ for $u > 0$ and set $\nu_p(0) = 0$. This finite convention makes the displayed exponents meaningful at step zero, where all positive-order differences vanish.

The valuation convention at zero is encoded by `KomorebiPaperIX.GlobalEulerBoundary.precision`. For p-adic congruences, divisibility is interpreted in $\mathbb{Z}_p$; statements over other coefficient rings specify the ring explicitly. Field divisibility in $\mathbb{Q}_p$ alone does not encode precision.

3.2 The standard address window and its common reduction

The addresses below the prime have unit diagonal coefficients. In this window the normalized packets are integral, and their reductions agree at every row.

Proposition 3.5 (Lucas reduction and address independence)**.** *Let p be prime, $m > 0$, and $N = mp$. For $0 \leq j < p$ and $n, q \in \mathbb{N}$,*

$$\binom{mpn + j}{mpq} \equiv \binom{mn}{mq} \pmod{p}.$$

Every $C_{mp,j}(n)$ belongs to $\mathbb{Z}_p$. Its reduction is the unique sequence $\overline{C}_m : \mathbb{N} \to \mathbb{F}_p$ satisfying

$$\sum_{q=0}^{n} \binom{mn}{mq} \overline{C}_m(q) = \delta_{n0} \quad \text{in } \mathbb{F}_p.$$

Consequently, for any two addresses $0 \leq j, k < p$,

$$C_{mp,j}(n) \equiv C_{mp,k}(n) \pmod{p} \qquad (n \in \mathbb{N}).$$

Proof. Lucas reduction separates the lowest base-p digits, which are j and zero in the upper and lower arguments. Their binomial contribution is one, leaving the displayed reduced row. On the diagonal the reduced coefficient is one. The recurrence (3.3) therefore constructs an integral packet by successive division by units in $\mathbb{Z}_p$. After reduction, every address satisfies the same triangular system over $\mathbb{F}_p$. Its diagonal entries are one, so its solution is unique. This proves both the common reduction and the congruence between addresses. The row reduction is `KomorebiPaperIX.StandardWindow.lucas_row_reduction`. The integral identification and the two reduction statements are `KomorebiPaperIX.WindowIndependence.standard_window_integral`, `KomorebiPaperIX.WindowIndependence.coeffPadic_coe`, `KomorebiPaperIX.WindowIndependence.residue_independence`, and `KomorebiPaperIX.WindowIndependence.two_window_addresses_congruent`. □

3.3 The full Nishibuchi statement and its starting bound

Theorem 3.6 (Nishibuchi congruence)**.** *Let p be prime, let $m > 0$, and suppose that $m \mid p-1$ or $m = 2$. Put $N = mp$ and $q = \operatorname{lcm}(2, p-1)$. For every $0 \le j < N$, all $U_{N,j}(n)$ belong to $\mathbb{Z}_p$. For $r \ge 1$ and $n \ge r$,*

$$E^{(N,j)}_{Nn+qp^r} \equiv E^{(N,j)}_{Nn} \pmod{p^r}. \tag{3.4}$$

Thus the positive sufficient starting index $n_0 = r$ works uniformly. If $m \mid p-1$, the stronger sufficient condition $r \le mn$ is valid.

This theorem covers the full parameter range and displacement of Conjecture 3.9 in the fixed version of Nishibuchi's paper used for the statement comparison. The corresponding Lean theorem is `KomorebiPaperIX.NishibuchiCompletion.conjecture_3_9`. The stronger divisor-branch threshold is `KomorebiPaperIX.NishibuchiCompletion.divisor_period`. The starting index is sufficient uniformly, although it need not be the earliest one for a given tuple of parameters.

We first prove the difference estimates on the divisor branch, then derive the period from a finite Newton expansion.

Theorem 3.7 (All-address differences on the divisor branch)**.** *Let p be prime, $m > 0$, $m \mid p-1$, $N = mp$, and $0 \le j < N$. Write $h = (p-1)/m$. For every $u, k, n \in \mathbb{N}$,*

$$U_{N,j}(n) \in \mathbb{Z}_p, \tag{3.5}$$

$$\Delta^k_{uh} U_{N,j}(n) \in p^{\min(k,mn)}\mathbb{Z}_p, \tag{3.6}$$

$$v_p(\Delta^k_{uh} C_{N,j}(n)) \ge \min(k, mn) - v_p(j!). \tag{3.7}$$

The factorial correction is part of the normalized statement, including addresses $j \ge p$.

Proof. Work in the Hurwitz integral ring of formal series

$$\mathcal{H}_p = \left\{ F \in \mathbb{Q}_p[[t]] : d![t^d]F \in \mathbb{Z}_p \text{ for every } d \right\}.$$

For $D = d/dt$, let $B_p = (D^{p^2} - D^p)/p$. Call F flat if $B_p^s F \in \mathcal{H}_p$ for every $s \ge 0$. We will prove that $j!\mathcal{P}_{N,j}$ is flat at every address. The main difficulty is to include the addresses at which the usual unit-pivot argument fails.

Set

$$H_c(t) = \sum_{n \ge 0} \frac{t^{Nn+c}}{(Nn+c)!}, \qquad A_c(t) = \sum_{n \ge 0} \frac{t^{Nn}}{(Nn+c)!}, \qquad R_c(t) = \frac{A_{c-1}(t)}{A_c(t)} \quad (1 \le c < N).$$

The denominator of this last quotient has nonzero constant coefficient. The section identities give

$$R_c H_c = t H_{c-1}, \qquad R_c(0) = c, \qquad j!\mathcal{P}_{N,j} = \mathcal{P}_{N,0} \prod_{c=1}^{j} R_c.$$

Cancellation of consecutive source sections proves the product identity and ties the ratios directly to the family under study.

At the singular addresses $c = ap$, $1 \le a < m$, use the weighted ring

$$\mathcal{W}_p = \left\{ \sum_{d \ge 0} a_d T^d : p^{\lfloor d/p \rfloor} a_d \in \mathbb{Z}_p \right\}.$$

The key estimate is $R_{ap}(\log(1+T)) \in \mathcal{W}_p$. Its arithmetic input comes from the cyclic group ring $\mathbb{Z}[\mathbb{Z}/N\mathbb{Z}]$. For its canonical element x and $I = (p, x^{p-1} - 1)$, the relation $(x^{p-1})^p = 1$ implies

$I^p \subseteq (p)I$. Induction yields $I^q \subseteq (p^{v_p(q!)})$. Grouping the falling factorial $(x)_d = \prod_{i=0}^{d-1}(x-i)$ into consecutive blocks of length p then gives an integral residue-coordinate factor $p^{v_p(\lfloor d/p \rfloor !)}$. The identity

$$v_p(d!) = \lfloor d/p \rfloor + v_p(\lfloor d/p \rfloor !)$$

accounts for precisely the weight in $\mathcal{W}_p$ after dividing by $d!$. Comparison of the section and group-ring recurrences identifies these coordinates with the logarithmically substituted section.

To pass from the section to the singular ratio, factor out its initial zero: for $c = ap$, the section has the form

$$H_c(\log(1+T)) = \frac{T^c}{c!}\,V(T), \qquad V(0) = 1, \qquad V \in \mathcal{W}_p.$$

The weight of $c!$ compensates for the coefficient shift by c. The weighted ring is closed under inversion of series with constant coefficient one and under $L_T = \log(1+T)(1+T)d/dT$. With $\ell(T) = \log(1+T)/T$, interpreted by coefficient shifting, differentiation of the displayed factorization gives

$$R_c(\log(1+T)) = c(1+T)\ell(T) + (L_T V)V^{-1}.$$

Since $p\ell(T) \in \mathcal{W}_p$ and $p \mid c$, both terms are weighted. This is the singular estimate `KomorebiPaperIX.SingularRatioAmice.paper_singular_amice_estimate`.

To deduce flatness from the weighted estimate, use $\Theta = (1+T)d/dT$. Both Θ and $(\Theta^{p^2} - \Theta^p)/p$ preserve $\mathcal{W}_p$. For the first operator, its coefficient formula is $(d+1)a_{d+1} + da_d$; whenever the floor weight rises, the factor $d+1$ provides the missing p. For the second operator, the proof uses its full Leibniz correction and the scalar congruence $p^2 \mid b^{p^2} - b^p$. Weighted monomials factor as

$$a_d T^d = (p^{\lfloor d/p \rfloor} a_d)(T^p/p)^{\lfloor d/p \rfloor} T^{d \bmod p}.$$

The operator preserves these monomials and hence finite weighted polynomials. Each fixed output coefficient depends on only finitely many input coefficients, which extends the result to the formal series. Substitution $T = e^t - 1$ intertwines D with Θ and maps $\mathcal{W}_p$ into $\mathcal{H}_p$, because

$$d![t^d]F(e^t - 1) = [T^0]\Theta^d F.$$

It follows that the singular ratios are flat. This transfer is proved in `KomorebiPaperIX.WeightedDworkTransfer.singular_ratio_flat`.

At a nonsingular address $2 \le c < N$ with $p \nmid c$, differentiation of the section identities gives

$$tR_c' = R_c(1 + R_{c-1} - R_c).$$

Writing $q = R_{c-1} - (c-1)$ and $y = R_c - c$, this becomes

$$(tD + c)y = cq + yq - y^2.$$

On degrees divisible by p, the inverse of $tD + c$ divides the degree-d coefficient by the unit $d + c$. Starting with $y_0 = 0$, form

$$y_{s+1} = (tD + c)^{-1}(cq + y_s q - y_s^2).$$

Flatness and support are preserved. Since the difference of two right sides factors as $(z - y)(q - z - y)$ and its second factor has zero constant term, agreement with the solution gains one coefficient at each iteration. Consequently $[t^d]y_s = [t^d]y$ for $d \le s$. Every coefficient of $B_p^r y$ can therefore be read from a sufficiently long finite iterate. The argument is entirely formal. This is the content of `KomorebiPaperIX.RiccatiFlatPropagation.riccati_solution_flat`.

The standard window supplies the initial ratio R_1. The weighted estimate handles $p \mid c$, and Riccati propagation handles $p \nmid c$. Flatness is closed under products, so the exact factorization establishes flatness of $j!\mathcal{P}_{N,j}$. Extracting its supported Hurwitz coefficients and applying the proved saturation and step-multiple lemmas yields (3.5)–(3.6). Finally $\Delta_{uh}^k U_{N,j} = j!\Delta_{uh}^k C_{N,j}$ gives (3.7), with the zero case retained. The resulting statements are `KomorebiPaperIX.AllResidueEuler.all_residue_flat`, `KomorebiPaperIX.AllResidueEuler.global_unnormalized_bound`, and `KomorebiPaperIX.AllResidueEuler.global_normalized_precision`. □

Example 3.8 (Why the factorial loss is necessary)**.** Take $p = 3$, $m = 2$, $N = 6$, and $j = 3$. The recurrence gives

$$C_{6,3}(1) = -\frac{1}{84}, \qquad C_{6,3}(2) = \frac{703}{5460}, \qquad C_{6,3}(2) - C_{6,3}(1) = \frac{64}{455}.$$

The last difference has 3-adic valuation zero. Here $h = 1$, so at $n = 1$ the order-one bound without factorial loss would require valuation at least one. The corrected normalized bound is sharp, since

$$v_3\big(C_{6,3}(2) - C_{6,3}(1)\big) = 0 = 1 - v_3(3!).$$

Multiplication by 3! gives

$$U_{6,3}(2) - U_{6,3}(1) = \frac{384}{455}, \qquad v_3\big(U_{6,3}(2) - U_{6,3}(1)\big) = 1,$$

consistent with the unnormalized theorem. Thus the factorial correction is already necessary at a single step in the all-address parameter range. These exact values and the refutation of a universal bound without loss are formalized in `KomorebiPaperIX.Normalization.normalized_first`, `KomorebiPaperIX.Normalization.normalized_second`, `KomorebiPaperIX.Normalization.normalized_difference`, `KomorebiPaperIX.Normalization.difference_valuation`, `KomorebiPaperIX.Normalization.factorial_loss_is_sharp`, `KomorebiPaperIX.Normalization.unnormalized_difference`, `KomorebiPaperIX.Normalization.unnormalized_difference_valuation`, and `KomorebiPaperIX.Normalization.universal_no_loss_refuted`.

Corollary 3.9 (A common period in the standard window)**.** *Let p be prime, $m > 0$, $m \mid p - 1$, and $0 \le j < p$. Put $h = (p-1)/m$. For every $n \ge 1$ and $u \in \mathbb{N}$,*

$$\overline{C}_m(n + uh) = \overline{C}_m(n), \qquad C_{mp,j}(n + uh) \equiv C_{mp,j}(n) \pmod{p}.$$

Thus h is a sufficient period from row one, common to all addresses in the standard window. It need not be the least period.

Proof. Since $j < p$, the factorial $j!$ is a unit in $\mathbb{Z}_p$. Also $mn \ge 1$. The order-one case of Theorem 3.7 therefore gives valuation at least one for the normalized difference at every step uh. This proves the congruence, and Proposition 3.5 identifies its reduction with the equality for $\overline{C}_m$. The preserved Lean development also proves this period directly from the Hurwitz inverse in characteristic p: `KomorebiPaperIX.WindowPeriod.uniform_standard_window_period`, `KomorebiPaperIX.WindowPeriod.multiple_period`, and `KomorebiPaperIX.WindowPeriod.standard_window_period_divisibility`. □

Lemma 3.10 (Prime-power period from finite differences)**.** *Let R be a commutative ring, p a prime, $f : \mathbb{N} \to R$, $h, n \in \mathbb{N}$, and $r \ge 1$. If*

$$p^{\min(k,r)} \mid \Delta_h^k f(n) \quad \text{for every } k \in \mathbb{N},$$

then $p^r \mid f(n + p^{r-1}h) - f(n)$.

Proof. The finite Newton expansion gives

$$f(n+p^{r-1}h)-f(n)=\sum_{k=1}^{p^{r-1}}\binom{p^{r-1}}{k}\Delta_h^k f(n).$$

Terms with $k\ge r$ have the required factor from the difference. For $1\le k<r$, the identity

$$k\binom{p^{r-1}}{k}=p^{r-1}\binom{p^{r-1}-1}{k-1}$$

and $v_p(k)<k$ imply $p^r\mid\binom{p^{r-1}}{k}p^k$ in $\mathbb{Z}$. Mapping that integer divisibility to R proves the claim term by term. The argument includes $r=1$. The declaration is `KomorebiPaperIX.PeriodFromDifferences.prime_power_period`. □

Proof of Theorem 3.6. On the divisor branch, $m\mid q$ and the packet step q/m is an integer multiple of $h=(p-1)/m$. If $r\le mn$, Theorem 3.7 supplies the hypotheses of Lemma 3.10. Its packet displacement is $p^{r-1}q/m$, which converts to the original degree as follows:

$$mp\left(n+p^{r-1}\frac{q}{m}\right)=mpn+qp^r.$$

The choice $n\ge r$ implies $mn\ge r$ because $m\ge 1$.

For an odd prime, $2\mid p-1$, so the option $m=2$ introduces no further case. The sole remaining branch is $p=2,m=2,N=4$. The all-row theorem proved next gives $2^k\mid\Delta_1^kC_{4,j}(n)$ at every row and address. After multiplying by $j!$, Lemma 3.10 gives packet period 2^{r-1} and degree displacement $4\cdot 2^{r-1}=2^{r+1}=qp^r$. It also gives integrality. Hence the complete statement follows. The formal degree conversion is `KomorebiPaperIX.NishibuchiCompletion.original_degree_shift`; the all-parameter degree theorem is `KomorebiPaperIX.NishibuchiCompletion.full_period_degree`. □

3.4 All-row dyadic differences and displacement amplification

For the remainder of the four-packet discussion abbreviate $C_j=C_{4,j}$ and $U_j=U_{4,j}$. Removing the original row cap requires an unshifted coefficient operator.

Theorem 3.11 (All-row dyadic higher differences)**.** *For $0\le j<4$ and all $u,k,n\in\mathbb{N}$,*

$$C_j(n)\in\mathbb{Z}_2, \tag{3.8}$$

$$\Delta_u^kC_j(n)\in 2^{k(1+\nu_2(u))}\mathbb{Z}_2, \tag{3.9}$$

$$\Delta_u^kU_j(n)\in 2^{k(1+\nu_2(u))+v_2(j!)}\mathbb{Z}_2. \tag{3.10}$$

There is no restriction $k\le n$ and no minimum with the row n.

Proof. The Hurwitz coefficients of the normalized source $K_{4,j}$ at degrees $4n$ are

$$g_j(n)=\binom{4n+j}{4n}^{-1}.$$

Their explicit expressions are

$$g_0(n)=1,\qquad g_1(n)=\frac{1}{4n+1},$$

$$g_2(n)=\frac{1}{(4n+1)(2n+1)},\quad g_3(n)=\frac{3}{(4n+1)(2n+1)(4n+3)}.$$

Every denominator is odd. An affine sequence with even slope satisfies $2^k \mid \Delta_1^k f(n)$ for every k, n. The exact product identity

$$\Delta_1^s(fg)(n) = \sum_{i=0}^{s} \binom{s}{i} \Delta_1^i f(n) \Delta_1^{s-i} g(n+i)$$

proves closure of this property under products. Applying the formula with the factors interchanged to $fg = 1$ isolates $f(n+s)\Delta_1^s g(n)$. Multiplication by $f(n+s)^{-1}$ and induction then prove closure under pointwise unit inverses. Thus all four g_j have the property. The explicit formulas and difference bounds are formalized in `KomorebiPaperIX.DyadicFourKernel.source_coe` and `KomorebiPaperIX.DyadicFourKernel.source_kummer`.

For an integer charge c, put

$$A_c = \frac{D^4 - c}{2}.$$

A series is flat of charge c when applying any power A_c^s to it gives an element of $\mathcal{H}_2$. The required product formula is

$$A_{c+d}(FG) = (A_c F)G + F(A_d G) + 2F'G''' + 3F''G'' + 2F'''G'. \tag{3.11}$$

All middle coefficients are integral. Derivatives preserve $\mathcal{H}_2$ and commute with A_c. Induction on the number of iterations in (3.11) proves that multiplication adds charges. If $FG = 1$, the same formula yields

$$A_{-c}G = -G^2 A_c F - G(2F'G''' + 3F''G'' + 2F'''G').$$

Induction, starting from the independently established integrality of the reciprocal G, proves that inversion negates the charge. Finally $A_{c+2} = A_c - 1$, so increasing a charge by two preserves flatness. These are `KomorebiPaperIX.ChargedDworkClosure.B_mul`, `KomorebiPaperIX.ChargedDworkClosure.flat_inverse`, and `KomorebiPaperIX.ChargedDworkClosure.flat_charge_shift`.

Packing g_j into degrees divisible by four gives $K_{4,j}$. Its source property is precisely charge-one flatness. Inversion gives charge -1, and the charge shift restores charge one for the reciprocal $\mathcal{P}_{4,j}$. If $\mathrm{dec}(F)(d) = d![t^d]F$, the exact coefficient identity is

$$\mathrm{dec}(A_1^s F)(d) = 2^{-s}\Delta_4^s \,\mathrm{dec}(F)(d).$$

At $d = 4n$, packing turns the right side into $2^{-s}\Delta_1^s C_j(n)$. This extraction requires no negative degree and therefore holds at every n, including zero. It proves the unit-step bound `KomorebiPaperIX.DyadicFourAllRows.base_step_bound`.

To incorporate $v_2(u)$, expand a displacement using finite Newton coefficients. For any prime p, $u > 0$, and $1 \le a \le u$,

$$p^{v_p(u)+1} \mid \binom{u}{a} p^a,$$

as follows from $a\binom{u}{a} = u\binom{u-1}{a-1}$ and $v_p(a) < a$. Repeated displacement, retaining the remaining base-difference order at each induction step, gives a factor $p^{k(1+v_p(u))}$ from k displacements. This induction is formalized in `KomorebiPaperIX.StepAmplification.repeated_precision`. Applying it with $p = 2$ proves (3.9); multiplication by $j!$ proves (3.10). Zero step and zero order follow from their definitions. The final statements are `KomorebiPaperIX.DyadicFourFinalStrengthening.normalized_step_bound` and `KomorebiPaperIX.DyadicFourFinalStrengthening.unnormalized_step_bound`. □

Corollary 3.12 (Step-sensitive divisor-branch bound)**.** *Under the hypotheses of Theorem 3.7, for all $u, k, n \in \mathbb{N}$,*

$$\Delta_{u(p-1)/m}^k U_{mp,j}(n) \in p^{\min(k(1+\nu_p(u)),mn)}\mathbb{Z}_p.$$

For $C_{mp,j}$ the corresponding valuation lower bound is $\min(k(1+\nu_p(u)), mn) - v_p(j!)$.

Proof. Apply the same repeated-displacement theorem with the fixed cap mn to (3.6). The argument preserves this cap and improves the contribution from the order of the difference. Division by the fixed factorial gives the rational valuation statement, with the zero case included. These are `KomorebiPaperIX.AllResidueStepAmplification.unnormalized_precision` and `KomorebiPaperIX.AllResidueStepAmplification.normalized_precision`. □

Corollary 3.13 (Dyadic periods from row zero)**.** *For $0 \le j < 4$, $r \ge 1$, and $n \ge 0$,*

$$U_j(n + 2^{r-1}) \equiv U_j(n) \pmod{2^r}.$$

For $j = 2, 3$ and $r \ge 2$, 2^{r-2} is also a sufficient packet period. The respective original degree displacements are 2^{r+1} and 2^r.

Proof. Use Lemma 3.10 and the unit-step case of Theorem 3.11. At $j = 2, 3$ the factorial contributes an additional power of two. The original degree versions are `KomorebiPaperIX.DyadicFourFinalStrengthening.all_rows_period_degree` and `KomorebiPaperIX.DyadicFourFinalStrengthening.all_rows_half_period_degree`. The next theorem sharpens these sufficient periods to a complete classification. □

3.5 Exact displacement valuations and least packet periods

Lemma 3.14 (Finite Newton dominance)**.** *Let $f : \mathbb{N} \to \mathbb{Q}$ and write $a_k = \Delta_1^k f(0)$. Suppose that $c \in \mathbb{N}$, $a_1 \ne 0$, $v_2(a_1) = c$, and*

$$v_2(a_k) \ge c + 1 + \lfloor \log_2 k \rfloor \qquad (k \ge 2).$$

Then for every $n \ge 0$ and $T > 0$,

$$f(n + T) - f(n) \ne 0, \qquad v_2\big(f(n + T) - f(n)\big) = v_2(T) + c.$$

Proof. Finite Newton reconstruction at an arbitrary row gives

$$\Delta_1^k f(n) = \sum_{i=0}^{n} \binom{n}{i} a_{k+i}.$$

The monotonicity of $\lfloor \log_2 k \rfloor$ therefore preserves the higher-coefficient bound after translation. In addition, $\Delta_1 f(n) - a_1 \in 2^{c+1}\mathbb{Z}_2$, so the first difference is nonzero of exact valuation c.

Now expand the displacement:

$$f(n + T) - f(n) = T\Delta_1 f(n) + \sum_{k=2}^{T} \binom{T}{k} \Delta_1^k f(n).$$

For $1 \le k \le T$, the binomial denominator identity implies

$$v_2\binom{T}{k} \ge v_2(T) - v_2(k) \ge v_2(T) - \lfloor \log_2 k \rfloor.$$

Every term of the remaining sum consequently has valuation at least $v_2(T) + c + 1$. Its sum cannot cancel the leading term, which has exact valuation $v_2(T) + c$. This proves nonvanishing and the claimed valuation using only finite sums. The general theorem is `KomorebiPaperIX.NewtonDominance.exact_displacement`. □

Theorem 3.15 (Exact four-packet displacement law)**.** *Set*

$$(c_0, c_1, c_2, c_3) = (1, 1, 4, 2), \qquad (d_0, d_1, d_2, d_3) = (1, 1, 5, 3).$$

For $0 \le j < 4$, $n \ge 0$, and every positive integer T,

$$\begin{aligned} C_j(n+T) - C_j(n) &\neq 0, & v_2\big(C_j(n+T) - C_j(n)\big) &= v_2(T) + c_j, \\ U_j(n+T) - U_j(n) &\neq 0, & v_2\big(U_j(n+T) - U_j(n)\big) &= v_2(T) + d_j. \end{aligned}$$

Proof. The first Newton coefficients, derived from (3.3), are

$$a_{0,1} = -2, \quad a_{1,1} = -\frac{6}{5}, \quad a_{2,1} = -\frac{16}{15}, \quad a_{3,1} = -\frac{36}{35}.$$

Their exact valuations are c_j. Theorem 3.11 gives $v_2(a_{j,k}) \ge k$ for all orders. To obtain the stronger logarithmic envelope required by Lemma 3.14, only a finite initial segment needs improvement. The recurrence proves the following exact valuations:

Address j	Further orders k	$v_2(a_{j,k})$ in that order
0	2	3
1	2	3
2	2, 3, 4, 5, 6	6, 9, 13, 16, 22
3	2, 3, 4	8, 8, 15

For all subsequent k, the elementary inequality $c_j + 1 + \lfloor \log_2 k \rfloor \le k$ completes the envelope. The entries follow by exact rational computation from the defining recurrence, and their valuations are proved in Lean. Together with the inequality for all subsequent orders, they establish the bound at every order. The uniform envelope is `KomorebiPaperIX.DyadicFourPrefix.tail_precision`.

Apply Lemma 3.14 to C_j. Multiplication by the nonzero factorial $j!$ adds $v_2(j!)$ and preserves nonzeroness. Hence $d_j = c_j + v_2(j!)$, giving the four valuation gains for the original coefficients. The declarations are `KomorebiPaperIX.DyadicFourLeastPeriods.normalized_exact_displacement` and `KomorebiPaperIX.DyadicFourLeastPeriods.exact_displacement`. □

Corollary 3.16 (All positive periods and their least element)**.** *For $r \in \mathbb{N}$ and $0 \le j < 4$, a positive integer T is a packet period of U_j modulo 2^r if and only if*

$$2^{\max(0, r-d_j)} \mid T.$$

Here a packet period means $U_j(n+T) \equiv U_j(n) \pmod{2^r}$ for every $n \ge 0$. Consequently the least positive packet periods are

j	0	1	2	3
least period	$2^{\max(0,r-1)}$	$2^{\max(0,r-1)}$	$2^{\max(0,r-5)}$	$2^{\max(0,r-3)}$.

Proof. By the exact displacement law, the congruence at any row is equivalent to $v_2(T) + d_j \ge r$. This is exactly the displayed integer divisibility criterion. Its least positive solution is the indicated power of two, which also covers $r = 0$ and $r \le d_j$. The all-period and least-element statements are `KomorebiPaperIX.DyadicFourLeastPeriods.period_iff_dvd` and `KomorebiPaperIX.DyadicFourLeastPeriods.least_packet_period`. □

For example, $U_2(n) \equiv 2 \pmod{32}$ and $U_3(n) \equiv 6 \pmod{8}$ for every row. At precision 2^{10} the least packet periods are $512, 512, 32, 128$. A packet displacement T has original degree displacement $4T$, as proved by `KomorebiPaperIX.DyadicFourLeastPeriods.original_degree_exa`

`ct_displacement`. Minimality here concerns packet shifts. Shifts of the full degree-indexed sequence, including its zero coefficients off the packet support, require a separate analysis. Exact valuations of arbitrary higher-order differences are also a separate question from the first-displacement law.

3.6 Classical odd-length coordinates and their congruence ladder

The classical signed Euler numbers have generating function $\operatorname{sech} t = \sum_{n\ge 0} E_{2n} t^{2n}/(2n)!$. They are the $N = 2$, $j = 0$ member of (3.1). The coordinate theorem reconstructs them from finite alternating sums of odd powers.

Definition 3.17 (Odd-length source and inverse coefficients)**.** For a positive odd integer p, define

$$S_d(p) = \sum_{a=0}^{p-1} (-1)^a (2a+1)^d.$$

Let γ_m be the integral Hurwitz coefficients of sech^2:

$$\operatorname{sech}^2 t = \sum_{m\ge 0} \gamma_m \frac{t^{2m}}{(2m)!}, \qquad \gamma_m = \sum_{b=0}^{2m} \binom{2m}{b} E_b E_{2m-b},$$

where $E_b = 0$ for odd b. Then $\gamma_0 = 1$. Define

$$T^{\mathrm{odd}}_{p,M}(n) = \sum_{m=0}^{\min(M,n)} \binom{2n}{2m} \gamma_m p^{2m} S_{2(n-m)}(p).$$

Here p need only be odd; primality is needed only to interpret the result as a prime-adic congruence ladder.

The source and inverse coefficients are `KomorebiPaperIX.OddPower.source` and `KomorebiPaperIX.EulerCoefficients.inverseEven`; the truncation is `KomorebiPaperIX.OddEuler.truncation`.

The same inverse coefficients can be written in the signed zigzag notation of the original manuscript. Define Z_r by

$$\sec t + \tan t = \sum_{r\ge 0} Z_r \frac{t^r}{r!}.$$

Then

$$\gamma_m = (-1)^m Z_{2m+1} \qquad (m \in \mathbb{N}). \tag{3.12}$$

Indeed, the odd coefficients come from $\tan t$, whose derivative is $\sec^2 t$. Changing the sign of the squared variable in the even series for $\operatorname{sech}^2 t$ gives $\sec^2 t$, so its degree-$2m$ Hurwitz coefficient is $(-1)^m \gamma_m$. Differentiation identifies this coefficient with Z_{2m+1}, proving the formula. The Lean bridge is `KomorebiPaperIX.Zigzag.inverse_is_signed_zigzag`; it uses the formal generating-function definition of the zigzag numbers.

Theorem 3.18 (Classical odd-length reconstruction)**.** *For every positive odd integer p and every $n \in \mathbb{N}$,*

$$E_{2n} = \sum_{m=0}^{n} \binom{2n}{2m} \gamma_m p^{2m} S_{2(n-m)}(p).$$

Equivalently, $E_{2n} = \sum_{q=0}^{n} V_{nq} S_{2q}(p)$ for the lower triangular matrix

$$V_{nq} = \binom{2n}{2q} \gamma_{n-q} p^{2(n-q)} \quad (q \le n), \qquad V_{nq} = 0 \quad (q > n).$$

It has $V_{nn} = 1$ and $p^{2(n-q)} \mid V_{nq}$ when $q \le n$.

Proof. The exponential generating series of the source is the finite sum

$$\mathcal{S}_p(t) = \sum_{a=0}^{p-1} (-1)^a e^{(2a+1)t}.$$

Multiplication by $\cosh t$ telescopes. Since p is odd,

$$\mathcal{S}_p(t) = \frac{1 + e^{2pt}}{2} \operatorname{sech} t.$$

Taking its even part gives

$$\sum_{n\geq 0} S_{2n}(p) \frac{t^{2n}}{(2n)!} = \cosh^2(pt) \operatorname{sech} t.$$

Multiplication by the reciprocal $\operatorname{sech}^2(pt)$ yields $\operatorname{sech} t$. The EGF product rule now gives the reconstruction coefficient by coefficient. Replacing m by $n - q$ produces the displayed lower triangular row. Its diagonal and weights follow from $\gamma_0 = 1$ and the explicit factor $p^{2(n-q)}$. The source-series identity is `KomorebiPaperIX.OddPower.odd_source_even_egf`; the reconstruction and row statement are `KomorebiPaperIX.OddEuler.odd_euler_reconstruction` and `KomorebiPaperIX.OddEuler.row_action`. □

Corollary 3.19 (Odd-prime truncation ladder)**.** *For every positive odd integer p and all $M, n \in \mathbb{N}$,*

$$p^{2M+2} \mid E_{2n} - T^{\mathrm{odd}}_{p,M}(n) \quad \text{in } \mathbb{Z}.$$

If $M \geq n$, the truncation equals E_{2n} exactly. In particular, for an odd prime p this is a congruence modulo p^{2M+2} in $\mathbb{Z}_p$.

Proof. Every omitted summand has $m \geq M+1$, hence contains the integer factor p^{2m} divisible by p^{2M+2}. All its other factors are integral. The finite sum of these omitted terms proves the assertion, with the empty-tail case giving exact equality. These are `KomorebiPaperIX.OddEuler.odd_euler_supercongruence` and `KomorebiPaperIX.OddEuler.full_depth_exact`. □

The reconstruction and its truncation bound already hold over the integers. For the dyadic construction we use a different source. Its original bound of 2^{2M+2} follows from integral reciprocal coefficients, and the next subsection sharpens that bound to the optimal uniform exponent $4M + 3$.

3.7 The optimal classical dyadic edge tower

Return now to the classical signed Euler numbers

$$\operatorname{sech} t = \frac{1}{\cosh t} = \sum_{n\geq 0} E_{2n} \frac{t^{2n}}{(2n)!}.$$

Thus $E_{2n} = E_{2n}^{(2,0)}$ in the convention (3.1). In particular $E_0 = 1$, $E_2 = -1$, and $E_4 = 5$. The following construction concerns these $N = 2$ coefficients, not $C_{4,0}$.

Definition 3.20 (Dyadic truncation)**.** Define β_m by the formal unit-series identity

$$\sum_{m\geq 0} \beta_m z^{2m} = \left(\frac{z}{\sinh z}\right)^2, \qquad \beta_0 = 1,$$

where the quotient means the inverse of the unit series $(\sinh z/z)^2$. Define

$$A(n) = \frac{3^{2n+2}-1}{4(2n+1)(2n+2)}, \qquad S_m(n) = \beta_m 2^{2m}\frac{(2n)!}{(2n-2m)!}A(n-m) \quad (0 \le m \le n),$$

and

$$T_M(n) = \sum_{m=0}^{\min(M,n)} S_m(n) \qquad (M, n \in \mathbb{N}). \tag{3.13}$$

These definitions correspond to `KomorebiPaperIX.Dyadic.beta`, `KomorebiPaperIX.Dyadic.sourceEven`, and `KomorebiPaperIX.DyadicReconstruction.truncation`. The formal summand uses a falling factorial, and `KomorebiPaperIX.DyadicReconstruction.falling_factorial_fraction` proves its equality with the quotient displayed above.

Lemma 3.21 (Exact reconstruction and source integrality)**.** *For every $n \in \mathbb{N}$,*

$$E_{2n} = \sum_{m=0}^{n} S_m(n).$$

Every β_m is 2-adically integral and every $A(n)$ is a nonzero 2-adic unit. If $M \ge n$, then $T_M(n) = E_{2n}$.

Proof. The ordinary kernel coefficients of $(\sinh z/z)^2$, in z^2, are

$$\frac{2^{2m+1}}{(2m+2)!}.$$

Legendre's strict factorial bound $v_2((2m+2)!) < 2m+2$ makes them integral. Their constant coefficient is one. The coefficient recurrence for its reciprocal therefore proves integrality of every β_m. For the source factor, lifting the exponent at the even power $2n+2$ gives

$$v_2(3^{2n+2}-1) = 2 + v_2(2n+2).$$

Its denominator has the same valuation, since $2n+1$ is odd. Thus $A(n)$ is a nonzero unit.

Its exponential generating function is

$$\mathcal{A}(t) = \sum_{n\ge 0} A(n)\frac{t^{2n}}{(2n)!} = \frac{\cosh(3t)-\cosh t}{4t^2}.$$

The quotients by powers of t are well defined because the required initial coefficients vanish. The hyperbolic identity

$$\mathcal{A}(t)\cosh t = \left(\frac{\sinh(2t)}{2t}\right)^2$$

then gives the exact formal-series product

$$\left(\sum_{m\ge 0}\beta_m(2t)^{2m}\right)\mathcal{A}(t) = \operatorname{sech} t.$$

Taking the factorial-scaled coefficient of t^{2n} produces the stated finite sum. Full depth leaves no omitted terms. The relevant declarations are `KomorebiPaperIX.Dyadic.beta_integral`, `KomorebiPaperIX.Dyadic.source_valuation`, `KomorebiPaperIX.DyadicSeries.dyadic_reconstruction_series`, and `KomorebiPaperIX.DyadicReconstruction.dyadic_reconstruction`. □

Lemma 3.22 (Exact weighted inverse coefficient)**.** *Let $\kappa_m = \beta_m 2^{2m}(2m)!$. For every $m \geq 1$,*

$$\kappa_m = -2^{4m}(2m-1)B_{2m} \neq 0, \qquad v_2(\kappa_m) = 4m-1,$$

where B_{2m} is the usual even Bernoulli number.

Proof. Let $Q(z) = z\coth z$, the regular formal series with constant coefficient one. The elementary differential identity

$$\left(\frac{z}{\sinh z}\right)^2 = Q(z) - zQ'(z)$$

and the Bernoulli expansion

$$Q(z) = 1 + \sum_{m\geq 1} \frac{2^{2m}B_{2m}}{(2m)!} z^{2m}$$

give $\beta_m = -2^{2m}(2m-1)B_{2m}/(2m)!$. Multiplication by $2^{2m}(2m)!$ proves the identity for κ_m. The Bernoulli valuation theorem gives $v_2(B_{2m}) = -1$. Since $2m-1$ is odd, the valuation is exactly $4m-1$. This also proves nonvanishing. In Lean, the hyperbolic identity is proved after clearing removable zeros. The library Bernoulli theorem then determines the valuation of κ_m. The declarations are `KomorebiPaperIX.DyadicSharpTower.kappa_bernoulli` and `KomorebiPaperIX.DyadicSharpTower.kappa_valuation`. □

Theorem 3.23 (Optimal uniform dyadic tower)**.** *For all $M, n \in \mathbb{N}$,*

$$E_{2n} - T_M(n) \in 2^{4M+3}\mathbb{Z}_2. \tag{3.14}$$

At the first omitted row $n = M+1$, the error is nonzero and

$$v_2\big(E_{2M+2} - T_M(M+1)\big) = 4M+3.$$

Thus the exponent $4M+3$ cannot be increased uniformly in the Euler row.

Proof. Rewrite the summand exactly as

$$S_m(n) = \kappa_m \binom{2n}{2m} A(n-m).$$

For $m > M$, its first factor is divisible by 2^{4m-1}, the binomial coefficient is integral, and $A(n-m)$ is a unit. Hence every omitted summand is divisible by 2^{4M+3}. Summing the finite tail proves (3.14). If $M \geq n$, the tail is empty and the statement includes the resulting zero error.

At $n = M+1$ there is exactly one omitted summand. Here $\binom{2n}{2n} = 1$ and $A(0) = 1$, so the error is κ_{M+1}. Lemma 3.22 gives its exact valuation $4M+3$ and nonvanishing. The statements are `KomorebiPaperIX.DyadicSharpTower.dyadic_supercongruence_sharp`, `KomorebiPaperIX.DyadicSharpTower.first_omitted_error`, and `KomorebiPaperIX.DyadicSharpTower.first_omitted_error_valuation`. □

Proposition 3.24 (Row-dependent summand weights)**.** *For $1 \leq m \leq n$, the summand is nonzero and*

$$v_2(S_m(n)) = 4m - 1 + v_2\binom{2n}{2m}.$$

More generally, if $L \in \mathbb{N}$ satisfies

$$L \le 4m - 1 + v_2\binom{2n}{2m} \qquad \text{for every } M < m \le n,$$

then $E_{2n} - T_M(n) \in 2^L\mathbb{Z}_2$.

Proof. Use the same factorization of $S_m(n)$, the nonzero binomial coefficient for $m \le n$, the exact valuation of κ_m, and the unit property of $A(n-m)$. The tail conclusion follows by addition in $2^L\mathbb{Z}_2$. These are `KomorebiPaperIX.DyadicSharpTower.summand_valuation` and `KomorebiPaperIX.DyadicSharpTower.dyadic_supercongruence_row_weight`. □

The minimum summand weight bounds the tail valuation from below, but cancellation may raise that valuation. An exact formula for the entire tail at every row therefore remains open. Uniform optimality follows from the equality already established at $n = M + 1$.

3.8 Isometric continuation of the four dyadic packets

The exact displacement law also determines the images of the packets after continuous extension to the dyadic integers.

Theorem 3.25 (Isometric packet bijections)**.** *For $0 \le j < 4$, define on natural numbers*

$$f_j(n) = \frac{U_j(n) - j!}{2^{d_j}}, \qquad (d_0, d_1, d_2, d_3) = (1, 1, 5, 3).$$

There is a unique continuous map $\Phi_j : \mathbb{Z}_2 \to \mathbb{Z}_2$ whose value at every $n \in \mathbb{N}$ equals the embedded rational $f_j(n)$. It is an isometric bijection and $\Phi_j(0) = 0$:

$$|\Phi_j(x) - \Phi_j(y)|_2 = |x - y|_2 \qquad (x, y \in \mathbb{Z}_2).$$

In particular, every element of $\mathbb{Z}_2$ has exactly one preimage.

Proof. For $n > 0$, the displacement law at row zero gives $v_2(U_j(n) - j!) = v_2(n) + d_j$, so $f_j(n) \in \mathbb{Z}_2$. The value at zero is zero. Applying the same law to the smaller of two unequal natural numbers shows

$$|f_j(n) - f_j(m)|_2 = |n - m|_2.$$

The natural numbers are dense in $\mathbb{Z}_2$, and the target is complete. The map therefore extends continuously. The distance identity extends on the dense product of natural-number points, making the extension an isometry. Density also proves uniqueness among all continuous maps with these prescribed values.

To prove surjectivity, use compactness. For an isometry g of a compact metric space into itself, the sequence of iterates $g^n(x)$ has a convergent subsequence. Two sufficiently close iterates, with exponents $a < b$, yield $d(x, g^{b-a}(x))$ equally small after removing a isometric iterations. Since $b - a > 0$, the latter point is in the image of g. Thus the image is dense. It is also compact and hence closed, so it is the whole space. Apply this to Φ_j on $\mathbb{Z}_2$.

The general ingredients are `KomorebiPaperIX.DensePadicIsometry.existsUnique_extensionEquiv` and `KomorebiPaperIX.CompactSelfIsometry.surjective`. The packet specialization is `KomorebiPaperIX.DyadicFourIsometry.existsUnique_isometry`; uniqueness among continuous maps is `KomorebiPaperIX.DyadicFourIsometry.unique_continuous`. □

Corollary 3.26 (Exact image of the original coefficients)**.** *The continuous extension of the original sequence is*

$$\widetilde{U}_j(x) = j! + 2^{d_j}\Phi_j(x).$$

It is injective, its image is exactly $j! + 2^{d_j}\mathbb{Z}_2$*, and*

$$|\widetilde{U}_j(x) - \widetilde{U}_j(y)|_2 = 2^{-d_j}|x - y|_2.$$

Every point of that affine image has exactly one preimage in $\mathbb{Z}_2$*.*

Proof. The formula agrees with U_j on every natural number. Continuity and the distance equality follow from the corresponding properties of Φ_j. Its surjectivity gives precisely the displayed affine image, and multiplication by the nonzero scalar 2^{d_j} preserves injectivity. The declarations are `KomorebiPaperIX.DyadicFourIsometry.originalExtension_nat`, `KomorebiPaperIX.DyadicFourIsometry.originalExtension_dist`, and `KomorebiPaperIX.DyadicFourIsometry.originalExtension_range`. □

The four rational packet sequences thus extend to bijections onto their exact dyadic images. The proof uses density, completeness, and the displacement law. Analyticity of these packet extensions would require a further argument. The analytic polynomial solution maps constructed later rest on separate hypotheses and a separate proof.

4 Finite obstructions, compatible lifts, and deformation

The arithmetic results above concern particular families. We now study the underlying correction problem, from finite polynomial lifts to continuous parameter charts and quantitative deformation. Throughout, we distinguish three objects: a residual obstruction, the set of correcting digits, and a root in a completion. A vanishing obstruction at one precision gives a lift to the next. Continuing indefinitely requires additional hypotheses.

Unless a different coefficient ring is specified, a polynomial system is a finite list of integer polynomials. All divisibility conditions on vectors are coordinatewise. For a positive integer q, write $S_q = \mathbb{Z}/q\mathbb{Z}$ and use an overline for reduction modulo q. For a prime p, write $\mathbb{F}_p = S_p$, normalize $|p|_p = p^{-1}$, and put

$$B_L(x) = x + p^L\mathbb{Z}_p^d.$$

An integer center is included in this notation by its natural image in $\mathbb{Z}_p^d$. All dimensions below are finite and may be zero. A statement over S_q allows composite q and $q = 1$ when explicitly stated. A cardinality formula in terms of vector-space rank uses prime p.

4.1 The polynomial correction equation

For $F = (F_1, \dots, F_e) \in \mathbb{Z}[X_1, \dots, X_d]^e$, define its evaluated Jacobian by the formal partial derivatives:

$$DF(x)_{ij} = (\partial_j F_i)(x), \qquad J_x = DF(x) \bmod q : S_q^d \longrightarrow S_q^e.$$

The obstruction therefore uses the Jacobian of the given polynomials. The following algebraic Taylor identity relates this linear map to polynomial evaluation.

Lemma 4.1 (Polynomial Taylor remainder)**.** *Let* R *be a commutative ring,* $P \in R[X_1, \dots, X_d]$*,* $a \in R^d$*, and* $c \in R$*. There is a polynomial* $Q \in R[U_1, \dots, U_d]$*, with* $Q(0) = 0$*, such that for every* $u \in R^d$*,*

$$P(a + cu) = P(a) + c\sum_{j=1}^{d}(\partial_j P)(a)u_j + c^2 Q(u). \tag{4.1}$$

In particular, when $R = \mathbb{Z}$ and a, u, c are integral, subtracting the first two terms on the right from $P(a + cu)$ gives a multiple of c^2.

Proof. Apply polynomial induction using constants, addition, and multiplication by a variable. Constants have zero remainder, and remainders add. For the multiplication step, suppose the assertion is known for P and write $D_P(U) = \sum_j (\partial_j P)(a)U_j$. Multiplication of its expansion by $a_i + cU_i$ gives constant term $P(a)a_i$, linear term $c(a_i D_P(U) + P(a)U_i)$, and remainder

$$c^2(Q(U)(a_i + cU_i) + D_P(U)U_i).$$

The product rule identifies the linear term with the derivative of PX_i. The displayed remainder vanishes at $U = 0$. This constructs the polynomial remainder before evaluation at u, even over rings where division by c would be invalid. □

The general polynomial identity is formalized by `Komorebi.PolynomialTaylorRing.exists_eval_scaled_remainder`. Its integer divisibility form is `Komorebi.PolynomialTaylor.taylor_dvd`.

Theorem 4.2 (Finite polynomial obstruction)**.** *Let $q \geq 1$, $L \geq 1$, $x \in \mathbb{Z}^d$, and suppose $F(x) \in q^L\mathbb{Z}^e$. Define the exact integer residual r_x by $F(x) = q^L r_x$, and put*

$$\mathcal{O}_x = S_q^e / \operatorname{im} J_x, \qquad o_x = [\overline{r_x}].$$

Then $\mathcal{O}_x$ is finite and

$$o_x = 0 \iff \exists y \in \mathbb{Z}^d : \quad y \equiv x \pmod{q^L}, \qquad F(y) \equiv 0 \pmod{q^{L+1}}. \tag{4.2}$$

No surjectivity or nonsingularity assumption is required.

Proof. Every point in the lower residue class has the form $y = x + q^L t$ with $t \in \mathbb{Z}^d$. By Lemma 4.1 and $2L \geq L + 1$,

$$F(x + q^L t) \equiv q^L(r_x + DF(x)t) \pmod{q^{L+1}}.$$

The integer q^L is nonzero. Cancelling it in the corresponding divisibility condition proves

$$F(x + q^L t) \equiv 0 \pmod{q^{L+1}} \iff \overline{r_x} + J_x \bar{t} = 0. \tag{4.3}$$

Integer vectors reduce onto S_q^d. The equation on the right is solvable exactly when $\overline{r_x}$ belongs to $\operatorname{im} J_x$, since that image is closed under negation. This is precisely $o_x = 0$. Finiteness follows because $\mathcal{O}_x$ is a quotient of the finite module S_q^e. □

The formal theorem `Komorebi.PolynomialObstruction.finite_polynomial_obstruction` presents the image using the integer-domain map $\mathbb{Z}^d \xrightarrow{DF(x)} \mathbb{Z}^e \to S_q^e$. Reduction onto S_q^d identifies that image with $\operatorname{im} J_x$. The class-preserving comparison is explicit in `Komorebi.PolynomialLiftCounts.obstructionEquiv`; the change of presentation does not change the distinguished residual class.

Proposition 4.3 (Affine depth zero)**.** *Let $F(x) = Ax - b$ for an integer linear map $A : \mathbb{Z}^d \to \mathbb{Z}^e$. The criterion of Theorem 4.2 holds for every $L \geq 0$, with $J = A \bmod q$.*

Proof. The affine identity $F(x + q^L t) = F(x) + q^L At$ has no quadratic remainder. The proof of (4.3) therefore needs no inequality involving $2L$. The residual and the finite quotient are obtained as above. □

The affine statement is formalized by `Komorebi.NativeAffine.finite_affine_obstruction`. Positive depth is essential in the nonlinear theorem: for $F(X) = X^2 - 1$, $q = 2$, $L = 0$, and $x = 0$, the Jacobian is zero and the residual class is nonzero, yet $y = 1$ is a root modulo two in the same residue modulo one.

Proposition 4.4 (Independence of the integer representative)**.** *Under the hypotheses of Theorem 4.2, let $x' = x + q^L s$ and write $F(x') = q^L r_{x'}$. Then*

$$J_{x'} = J_x, \qquad \overline{r_{x'}} = \overline{r_x} + J_x \overline{s}.$$

Consequently the two residual classes coincide in the common quotient $S_q^e / \operatorname{im} J_x$.

Proof. Polynomial evaluation respects coordinatewise congruence. Apply this first to every formal partial derivative: $x' \equiv x \pmod q$ gives $J_{x'} = J_x$. Next apply (4.1) to F at scale q^L. Subtract the two exact residual identities and cancel q^L to obtain the displayed relation modulo q. Its additional term belongs to the common Jacobian image, so it vanishes in the quotient. □

This comparison is formalized by `Komorebi.PolynomialRepresentative.representative_independence`. This proves independence of the integer representative within a fixed polynomial presentation. It makes no identification of the raw cokernels of different equation lists. For example, adjoining a zero equation leaves the roots unchanged while adding an S_q summand to the raw cokernel.

4.2 One correction shared by two systems

The next obstruction concerns compatibility between individually attainable corrections. Its algebraic form works over an arbitrary commutative ring. Let U, V, W be R-modules and let $J : U \to V$, $K : U \to W$ be linear. Define

$$C(J, K) = U/(\ker J + \ker K). \tag{4.4}$$

An attainable value Ju has label $[u] \in C(J, K)$; this is independent of the choice of its preimage because $\ker J$ is killed in the quotient. Similarly, Kv has label $[v]$. Denote these maps by $\lambda_J : \operatorname{im} J \to C(J, K)$ and $\lambda_K : \operatorname{im} K \to C(J, K)$.

Theorem 4.5 (Common correction and fiber product)**.** *For every $u, v \in U$,*

$$[u] = [v] \text{ in } C(J, K) \quad \Longleftrightarrow \quad \exists w \in U : \ Jw = Ju, \quad Kw = Kv. \tag{4.5}$$

Both label maps are surjective, and the image of the joint map is exactly the fiber product:

$$\operatorname{im}\big(U \to \operatorname{im} J \times \operatorname{im} K, \ u \mapsto (Ju, Ku)\big) = \operatorname{im} J \times_{C(J,K)} \operatorname{im} K. \tag{4.6}$$

If either V or W is finite, then $C(J, K)$ is finite.

Proof. If $[u] = [v]$, write $u - v = s + t$ with $s \in \ker J$ and $t \in \ker K$. Then $w = u - s = v + t$ satisfies both required equalities. Conversely, a common w gives

$$u - v = (u - w) + (w - v) \in \ker J + \ker K.$$

Every class $[u]$ is the label of Ju and of Ku, proving surjectivity. For attainable $a = Ju$ and $b = Kv$, the first equivalence says exactly that equal labels are equivalent to a simultaneous preimage. This proves (4.6). Finally, a finite target has finite image, and the associated surjective label map makes $C(J, K)$ finite. □

These statements are `Komorebi.SharedCorrection.quotient_mk_eq_iff_common` and `Komorebi.SharedCorrection.joint_range_eq_fiber`, together with the finite-target corollaries in the same module. The quotient also has the expected universal property: every linear map from U annihilating both kernels factors uniquely through $C(J, K)$. Equivalently, compatible linear observations on $\operatorname{im} J$ and $\operatorname{im} K$ factor uniquely through their label maps. This is formalized by `Komorebi.SharedCorrection.common_paired_universal`.

Corollary 4.6 (Simultaneous polynomial lifting)**.** *Let $F \in \mathbb{Z}[X_1, \ldots, X_d]^e$ and $G \in \mathbb{Z}[X_1, \ldots, X_d]^f$ have a common lower root x modulo q^L, where $q \geq 1$ and $L \geq 1$. Use the integer-domain reduced Jacobians $J : \mathbb{Z}^d \to S_q^e$ and $K : \mathbb{Z}^d \to S_q^f$. Suppose $u, v \in \mathbb{Z}^d$ are separately valid corrections for F and G, respectively. Then*

$$[u] = [v] \text{ in } \mathbb{Z}^d/(\ker J + \ker K)$$

if and only if a single $y \equiv x \pmod{q^L}$ solves both systems modulo q^{L+1}. Each separate label is independent of the chosen valid correction. The common quotient is finite.

Proof. For each system, (4.3) translates a valid next root into a linear correction equation. Once u corrects F, another digit w corrects F precisely when $Jw = Ju$. The analogous condition for G is $Kw = Kv$. Theorem 4.5 now gives the criterion. Two valid corrections of one block differ by a vector in its Jacobian kernel, so they have equal labels. Finiteness follows from either finite residue target. An unconditional formulation quantifies over separately valid u, v with equal labels, thereby including their existence in the criterion rather than assuming it. □

See `Komorebi.PolynomialShared.joint_lift_iff_class_eq` and `Komorebi.PolynomialShared.finite_polynomial_pair_compatibility`.

For affine integer systems $F(X) = AX - b$ and $G(X) = BX - c$, where $A : \mathbb{Z}^d \to \mathbb{Z}^e$ and $B : \mathbb{Z}^d \to \mathbb{Z}^f$ are linear, Corollary 4.6 holds for $q \geq 1$ and every $L \geq 0$, including depth zero. Here x is a common root modulo q^L, and J and K are the reductions of A and B with domain $\mathbb{Z}^d$. Given separately valid corrections u, v, a simultaneous next lift exists if and only if their classes in $\mathbb{Z}^d/(\ker J + \ker K)$ agree. The quotient is finite, and each label is independent of the chosen valid correction. Indeed, Proposition 4.3 supplies the correction equation without a positive-depth restriction, and the proof of Theorem 4.5 then applies unchanged. The formal statement is `Komorebi.AffinePair.joint_lift_iff_class_eq`. Quantifying over separately valid corrections with equal labels also includes their existence in the criterion, as in `Komorebi.AffinePair.finite_affine_pair_compatibility`. Separate liftability alone is insufficient. At $x = 0$ modulo two, $F(X) = X$ and $G(X) = X - 2$ both lift to modulus four, but their correcting digits are respectively 0 and 1 modulo two. No single residue solves both equations modulo four.

4.3 Correction torsors and exact residue counts

Integer representatives and residue classes at the next level must also be distinguished. For $q \geq 1$ and $L \geq 0$, the map

$$S_q^d \longrightarrow \{y \in (\mathbb{Z}/q^{L+1}\mathbb{Z})^d : y \equiv x \pmod{q^L}\}, \qquad t \longmapsto x + q^L \widetilde{t}$$

is a bijection, where $\widetilde{t}$ is any integer representative. Independence of the representative and injectivity both follow by cancelling q^L in integer divisibility; surjectivity follows by expressing the lower congruence as a multiple of q^L. This is the equivalence `Komorebi.ResidueLiftDigits.digitEquivResidue`.

At positive polynomial depth, restrict it to the correction fiber

$$\mathcal{T}_x = \{t \in S_q^d : \overline{r_x} + J_x t = 0\}$$

and the lift set

$$\mathcal{L}_{x,L} = \{y \in (\mathbb{Z}/q^{L+1}\mathbb{Z})^d : y \equiv x \pmod{q^L},\ F(y) = 0\}.$$

The polynomial predicate in the second display is evaluation in the residue ring. Evaluation on canonical integer representatives gives the same predicate, because integer polynomial evaluation respects congruence.

Theorem 4.7 (Torsor and lift cardinalities)**.** *Under the hypotheses of Theorem 4.2, there is an equivalence $\mathcal{T}_x \simeq \mathcal{L}_{x,L}$. Every nonempty fiber is a torsor for* $\ker J_x$ *under addition of correction digits. If $q = p$ is prime and $\rho_x = \operatorname{rank}_{\mathbb{F}_p} J_x$, then*

$$\#\mathcal{L}_{x,L} = \begin{cases} p^{d-\rho_x}, & o_x = 0, \\ 0, & o_x \neq 0, \end{cases} \qquad \#\mathcal{O}_x = p^{e-\rho_x}. \tag{4.7}$$

Proof. Equation (4.3) identifies exactly which points of the residue fiber satisfy the equations, proving the restricted equivalence. If $u, t \in \mathcal{T}_x$, subtraction gives $J_x(t-u) = 0$. Conversely, adding any element of $\ker J_x$ preserves the correction equation. The unique kernel vector carrying u to t is $t - u$, so the action is free and transitive. Choosing one valid digit identifies the fiber with the kernel, without making that choice part of its translation action.

For prime p, rank-nullity gives $\dim \ker J_x = d - \rho_x$ and $\dim(\mathbb{F}_p^e / \operatorname{im} J_x) = e - \rho_x$. The cardinality of an n-dimensional $\mathbb{F}_p$-space is p^n. Combine these identities with the obstruction criterion and the residue equivalence. □

The cardinality formula is formalized by `Komorebi.PolynomialLiftCounts.finite_polynomial_lift_count`. The explicit torsor and chosen-point equivalence are developed in `Komorebi.FiniteFiberCounts.translate_exists_unique` and `Komorebi.PolynomialLiftCounts.liftEquivKernel`. For composite q, the torsor and kernel cardinality remain valid, while the vector-space rank formula is unavailable. Multiplication by two on $\mathbb{Z}/4\mathbb{Z}$ already has a kernel of size two.

4.4 Smooth compatible towers and exact roots

Surjectivity removes every residual obstruction and persists throughout a positive-depth residue ball. This supplies the additional hypothesis needed to iterate the finite theorem.

Theorem 4.8 (Compatible polynomial tower)**.** *Let $q \geq 1$, $L \geq 1$, $F \in \mathbb{Z}[X_1, \ldots, X_d]^e$, and $x \in \mathbb{Z}^d$ satisfy $F(x) \in q^L\mathbb{Z}^e$. Suppose $J_x : S_q^d \to S_q^e$ is surjective. There is a single sequence $z_n \in \mathbb{Z}^d$ such that*

$$z_0 = x, \qquad F(z_n) \in q^{L+n}\mathbb{Z}^e,$$
$$z_{n+1} - z_n \in q^{L+n}\mathbb{Z}^d, \qquad J_{z_n} = J_x.$$

For $a \leq b$, it also satisfies $z_b - z_a \in q^{L+a}\mathbb{Z}^d$.

Proof. At level $N = L + n$, divide the residual by q^N. Surjectivity supplies $t \in S_q^d$ with $J_{z_n} t = -\overline{F(z_n)/q^N}$. Choose an integer representative and apply (4.3) to obtain z_{n+1}. The new point is congruent to z_n modulo q. Evaluation of each formal partial derivative therefore gives $J_{z_{n+1}} = J_{z_n}$.

Construct the sequence recursively, retaining the root condition and the fixed Jacobian at each step. Surjectivity at every later stage then follows from the single initial assumption. The cross-level congruence follows by induction, adding consecutive differences whose divisibilities all imply divisibility by q^{L+a}. □

The full tower statement is `Komorebi.PolynomialIntegerTower.exists_compatible_tower_strong`. The proof constructs a compatible path using classical choice. It does not claim an optimized procedure for selecting its digits.

Theorem 4.9 (Completion and smooth branching)**.** *In Theorem 4.8, let $q = p$ be prime. The tower can be chosen with a limit $a \in \mathbb{Z}_p^d$ such that*

$$F(a) = 0, \qquad z_n \longrightarrow a, \qquad a - z_n \in p^{L+n}\mathbb{Z}_p^d \quad (n \geq 0).$$

At each stage, the number of distinct next-level residue roots over z_n is exactly p^{d-e}, and the finite obstruction module has one element. If $d = e$, there is exactly one root of F in $B_L(x)$.

Proof. Cross-level compatibility bounds the supremum distance between later terms by $p^{-(L+n)}$. Completeness gives a limit, and closed norm bounds retain the congruence to every z_n. View the coefficients of the original integer polynomials in $\mathbb{Z}_p$. Their evaluations satisfy both $F_i(z_n) \to F_i(a)$, by continuity, and $F_i(z_n) \to 0$, by their growing divisibility. Uniqueness of limits gives exact vanishing.

Surjectivity gives $\operatorname{rank} J_x = e \leq d$. Theorem 4.7 then gives p^{d-e} next residue roots and obstruction cardinality one at every stage.

For square uniqueness, let $a, b \in B_L(x)$ be arbitrary exact roots. Their canonical integer truncations are genuine roots in the respective finite residue rings, since evaluation commutes with reduction. They agree at depth L. If they agree at depth $N \geq L$, both next residues belong to the same lift set, which has cardinality $p^{d-d} = 1$. Induction gives agreement at every depth. The residue maps separate p-adic integers, so $a = b$. Existence was already supplied by the tower. □

The endpoint construction is `Komorebi.PolynomialPadicLift.exists_padic_root_with_tower`, the branching statement is `Komorebi.PolynomialSmoothCounts.exists_regular_tower`, and uniqueness for all roots in the ball is `Komorebi.PolynomialPadicUnique.exists_unique_padic_root`.

Remark 4.10. The one-step singular criterion and the smooth infinite theorem have different hypotheses. The polynomial $X^2 - 5$ has odd roots modulo two and four but has no root modulo eight. Thus a previous zero obstruction does not force the next one to vanish. Also, rectangular surjectivity alone does not imply uniqueness: $F(X, Y) = X$ has the distinct dyadic roots $(0, 0)$ and $(0, 2)$ in the same residue ball modulo two. These examples show why the hypotheses of the tower theorem and its uniqueness clause are needed.

4.5 Coefficient stability and continuous root charts

Proposition 4.11 (Smooth coefficient stability)**.** *Let p be prime, $M \geq 1$, $F, H \in \mathbb{Z}[X_1, \ldots, X_d]^e$, and $c \in p^M\mathbb{Z}$. Suppose $DF(x) \bmod p$ is surjective for an integer $x \in \mathbb{Z}^d$. If $a \in \mathbb{Z}_p^d$ satisfies $F(a) = 0$ and $a \equiv x \pmod p$, then there is a root b of $G = F + cH$ with $b - a \in p^M\mathbb{Z}_p^d$.*

If $d = e$ and $M \geq L \geq 1$, every pair of roots $a, b \in B_L(x)$ of F and G, respectively, satisfies $b - a \in p^M\mathbb{Z}_p^d$. The perturbed root is unique in that ball whenever it exists.

Proof. Since $p \mid c$, differentiation gives $DG(x) \equiv DF(x) \pmod{p}$. Truncate a to an integer vector y at depth M. Then $F(y) \equiv 0 \pmod{p^M}$, hence also $G(y) \equiv 0 \pmod{p^M}$. Congruence of y and x modulo p preserves the reduced Jacobian. The smooth root theorem gives $b \equiv y \equiv a \pmod{p^M}$.

In the square case this constructed root lies in $B_L(x)$ because $M \geq L$. Square uniqueness for G identifies it with every competing root in that ball. The construction therefore gives the precision assertion for an arbitrary root there. □

See `Komorebi.PolynomialRootStability.exists_nearby_perturbed_root` and `Komorebi.PolynomialRootStability.perturbed_roots_congruent`. The scalar c may be negative or zero. The weighted theorems below allow arbitrary integral p-adic coefficients and give an explicit first-order displacement.

Theorem 4.12 (Continuous charts with a selected Jacobian minor)**.** *Let p be prime, $L \geq 1$, $F \in \mathbb{Z}[X_1, \ldots, X_{e+k}]^e$, and $x \in \mathbb{Z}^{e+k}$ satisfy $F(x) \in p^L\mathbb{Z}^e$. Suppose the full reduced Jacobian is surjective. There is a permutation σ of the coordinates such that, writing $\sigma x = (y_0, t_0)$ with e solved coordinates and k parameter coordinates, the parameter ball $T = t_0 + p^L\mathbb{Z}_p^k$ is homeomorphic to the entire root locus*

$$\mathcal{R} = \{z \in B_L(x) : F(z) = 0\}.$$

The inverse homeomorphism is the actual free-coordinate projection $z \mapsto \mathrm{free}(\sigma z)$. If $h : T \to \mathcal{R}$ is the forward map, then for every $N \geq 0$,

$$h(t) \equiv h(u) \pmod{p^N} \iff t \equiv u \pmod{p^N}. \tag{4.8}$$

Equivalently, $\sigma h(t) = (\phi(t), t)$ for a continuous $\phi : T \to y_0 + p^L\mathbb{Z}_p^e$ whose graph is the entire reordered root locus.

Proof. First consider a square integer polynomial map P whose reduced Jacobian at x is surjective. Its Jacobian is invertible over $\mathbb{F}_p$. A digit comparison using Taylor expansion shows that equal outputs modulo p^{N+1} force equal inputs modulo p^{N+1} whenever the inputs already agree modulo p^N and belong to $B_1(x)$. Induction and integer truncation give, for arbitrary $a, b \in B_1(x)$,

$$P(a) \equiv P(b) \pmod{p^N} \iff a \equiv b \pmod{p^N}.$$

More generally, for a rectangular map with surjective reduced Jacobian,

$$P(B_L(x)) = P(x) + p^L\mathbb{Z}_p^e. \tag{4.9}$$

One inclusion follows from polynomial congruence. For integer targets in the right-hand ball, subtract the target from the constant terms and use the smooth root theorem. For arbitrary p-adic targets, approximate by integer truncations. The input ball is compact, so its continuous image is closed. It contains all these approximants and hence their limit. This argument does not assume that independently chosen preimages form a compatible sequence.

For square P, the resulting bijection between the two compact residue balls is continuous, and its target is Hausdorff. It is therefore a homeomorphism, with the same congruence preservation for its inverse.

Now suppose a chosen set of e columns of $DF(x) \bmod p$ is independent, and put those coordinates first. Apply the square result to the augmented polynomial map

$$\Psi(Y, T) = (F(Y, T), T).$$

Its reduced Jacobian has upper blocks $(D_Y F, D_T F)$ and lower blocks $(0, I_k)$, so it is invertible. For $t \in T$, the target $(0, t)$ belongs to $\Psi(x) + p^L \mathbb{Z}_p^{e+k}$ because $F(x) \equiv 0 \pmod{p^L}$. The inverse of Ψ at that target gives the unique root with parameters t. The inverse congruence property gives one direction of (4.8); projection onto the parameter coordinates gives the other.

Finally, surjectivity of the original rectangular Jacobian guarantees a basis among its columns. Extend the corresponding injection of coordinate indices to a permutation σ. Renaming the variables by σ^{-1} transports polynomial evaluation and every formal partial derivative. The fixed-coordinate construction therefore applies to the original system after undoing the permutation. □

The main theorem is `Komorebi.PolynomialGeneralChart.exists_parameter_homeomorph`. The square local inverse and surjectivity for arbitrary targets are `Komorebi.PolynomialLocalInverse.localHomeomorph` and `Komorebi.PolynomialLocalSurjective.exists_padic_preimage`. The center x is only an approximate root: the chart need not satisfy $\phi(t_0) = y_0$ unless $F(x) = 0$ exactly. The assertion selects suitable coordinates; it does not permit an arbitrary prescribed solved block.

4.6 Prescribed parameters and their obstruction

Suppose now that the split into solved variables and parameter variables is fixed in advance. Let

$$F \in \mathbb{Z}[Y_1, \dots, Y_m, T_1, \dots, T_k]^e, \qquad x = (y_0, t_0),$$

and assume $F(x) = q^L r_x$ with $q \geq 1$ and $L \geq 1$. Write the reduced Jacobian as

$$DF(x) \bmod q = (J_x \; K_x), \qquad J_x : S_q^m \to S_q^e, \quad K_x : S_q^k \to S_q^e.$$

For a prescribed parameter digit $v \in S_q^k$, let $\mathcal{L}_{x,L,v}$ be the set of next residue roots over x whose parameter coordinates are $t_0 + q^L \widetilde{v}$ modulo q^{L+1}.

Theorem 4.13 (Prescribed-parameter obstruction and count)**.** *Define*

$$o_x(v) = [\overline{r_x} + K_x v] \in S_q^e / \operatorname{im} J_x.$$

Then $\mathcal{L}_{x,L,v}$ is nonempty if and only if $o_x(v) = 0$. When nonempty, it is a torsor for $\ker J_x$*. If $q = p$ is prime and $\rho_x = \operatorname{rank}_{\mathbb{F}_p} J_x$, then*

$$\#\mathcal{L}_{x,L,v} = \begin{cases} p^{m-\rho_x}, & o_x(v) = 0, \\ 0, & o_x(v) \neq 0, \end{cases} \qquad \#(\mathbb{F}_p^e / \operatorname{im} J_x) = p^{e-\rho_x}. \tag{4.10}$$

The criterion holds uniformly for every prescribed v without a surjectivity hypothesis.

Proof. A candidate for the next lift has the form $x + q^L(\widetilde{u}, \widetilde{v})$, with solved digit $u \in S_q^m$. Its Taylor correction equation is

$$\overline{r_x} + J_x u + K_x v = 0.$$

Solvability is exactly vanishing of the displayed cokernel class. Restrict the residue-digit equivalence to the prescribed parameter coordinates. This identifies the lift fiber with the solved-digit fiber. Translation by $\ker J_x$ is free and transitive on any nonempty such fiber. The prime-field count follows from rank-nullity as before. □

The formal statements start from the lower root and construct its residual: `Komorebi.PolynomialPrescribedObstruction.finite_prescribed_obstruction` and `Komorebi.PolynomialPrescribedObstruction.finite_prescribed_lift_count`. For $F(Y, T) = T$ at the dyadic origin, the full

Jacobian (0 1) is surjective but $J_x = 0$. Prescribing $v = 1$ forces $T = 2 \pmod 4$, so no root exists; prescribing $v = 0$ gives two choices of Y modulo four over the initial zero residue. This example explains why the criterion depends on the solved block, rather than merely on the full Jacobian.

Theorem 4.14 (Every compatible integer parameter path)**.** *In the preceding notation, assume J_x is surjective. For every supplied integer path $t_n \in \mathbb{Z}^k$ satisfying*

$$t_0 = \mathrm{free}(x), \qquad t_{n+1} - t_n \in q^{L+n}\mathbb{Z}^k,$$

there is a compatible polynomial tower $z_n \in \mathbb{Z}^{m+k}$ with

$$z_0 = x, \quad \mathrm{free}(z_n) = t_n, \quad F(z_n) \in q^{L+n}\mathbb{Z}^e, \quad z_{n+1} - z_n \in q^{L+n}\mathbb{Z}^{m+k}.$$

The free-coordinate equality is exact in integers. Both reduced Jacobian blocks remain fixed, and $z_b - z_a \in q^{L+a}\mathbb{Z}^{m+k}$ whenever $a \le b$.

Proof. At level $N = L + n$, write the supplied parameter increment exactly as $t_{n+1} - t_n = q^N w$ with $w \in \mathbb{Z}^k$. Surjectivity of the solved block gives a digit u satisfying $J_{z_n} u = -\overline{F(z_n)/q^N} - K_{z_n}\overline{w}$. Choose an integer representative $\widetilde{u}$ and set $z_{n+1} = z_n + q^N(\widetilde{u}, w)$. Its parameter coordinates equal the supplied target exactly. Taylor expansion proves the next root condition, while congruence modulo q preserves the full reduced Jacobian. Construct the sequence recursively, preserving the exact parameter coordinates and the fixed solved Jacobian. The earlier cross-level argument proves all the stated integer congruences. □

This is `Komorebi.PolynomialPrescribedIntegerTower.exists_prescribed_tower_strong`. In particular, no assumption about solvability at each later level has been added to the initial surjectivity condition.

Theorem 4.15 (Arbitrary p-adic parameter endpoints)**.** *Let $q = p$ be prime in Theorem 4.14. For every $\tau \in t_0 + p^L\mathbb{Z}_p^k$, there are an integer tower z_n and $b \in \mathbb{Z}_p^{m+k}$ such that*

$$F(b) = 0, \qquad \mathrm{free}(b) = \tau, \qquad z_n \longrightarrow b, \qquad b - z_n \in p^{L+n}\mathbb{Z}_p^{m+k}.$$

The initial value is $z_0 = x$. At every stage and for every proposed parameter digit $v \in \mathbb{F}_p^k$, the corresponding next residue fiber has exactly p^{m-e} elements. If $m = e$, then for each τ there is exactly one such root in $B_L(x)$, independently of all choices of integer approximants and correcting digits.

Proof. Set the first integer parameter approximation equal to the prescribed t_0. For $n \ge 1$, use canonical representatives of τ mod p^{L+n}. The initial congruence and truncation precision give

$$\tau - t_n \in p^{L+n}\mathbb{Z}_p^k.$$

Subtracting consecutive relations gives compatibility in $\mathbb{Z}_p$. Faithfulness of integer divisibility under $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$ gives compatibility in $\mathbb{Z}$, and the same norm bounds give $t_n \to \tau$. Theorem 4.14 and completion now produce b with exact polynomial vanishing and all retained precisions. Each free coordinate has both the corresponding coordinate of b and that of τ as its limit, proving exact parameter equality.

The solved Jacobian remains surjective, so every shifted obstruction $[\overline{r} + K_{z_n} v]$ vanishes. Formula (4.10) therefore gives p^{m-e} choices for every parameter digit at every stage. When $m = e$, the fixed-coordinate chart in the proof of Theorem 4.12 contains every root in the initial ball with the specified parameters. Its uniqueness proves the endpoint assertion. This is uniqueness of the actual endpoint, not of the integer representatives used to approach it. □

The endpoint theorems are `Komorebi.PolynomialPrescribedPadicTower.exists_prescribed_padic_root_with_tower` and `Komorebi.PolynomialPrescribedPadicTower.exists_unique_prescribed_padic_root`. The separate supplied-path theorem `Komorebi.PolynomialPrescribedPadicTower.exists_root_with_prescribed_path` accepts any compatible integer parameter path with limit τ. For $m > e$, endpoint uniqueness is false in general: the system $F(Y_1, Y_2, T) = Y_1$ has two distinct roots $(0, 0, 0)$ and $(0, 2, 0)$ in the same dyadic ball with identical parameter value.

4.7 Weighted lifting for a nonunit Jacobian

The remaining theorems in this section allow arbitrary coefficients and centers in $R = \mathbb{Z}_p$. They concern square solved Jacobians invertible over $\mathbb{Q}_p$, even when their reductions modulo p are singular. The exponent controlling an integral inverse is retained explicitly.

Theorem 4.16 (Weighted nonlinear lifting)**.** *Let p be prime, $d, s, r \in \mathbb{N}$, $F \in R[X_1, \dots, X_d]^d$, and $a \in R^d$. Set $J = DF(a)$. Suppose*

$$J \in \mathrm{GL}_d(\mathbb{Q}_p), \qquad p^s J^{-1} \in M_d(R), \qquad F(a) \in p^r R^d, \qquad r > 2s. \tag{4.11}$$

Then there is exactly one root b of F in $a + p^{s+1}R^d$, and it satisfies

$$b - a \in p^{r-s} R^d. \tag{4.12}$$

Uniqueness holds throughout the full ball of depth $s + 1$.

Proof. Let $A = p^s J^{-1} \in M_d(R)$. Inclusion into $\mathbb{Q}_p$ and its injectivity give the two integral identities

$$AJ = JA = p^s I_d.$$

These identities are also an equivalent integral certificate for the first two hypotheses of (4.11). In particular, A is injective: $Aw = 0$ implies $p^s w = JAw = 0$, hence $w = 0$.

Precondition the polynomial system by A and set $c = p^{s+1}$. Its derivative at a is $AJ = p^s I_d$. Lemma 4.1 gives a polynomial vector Q with $Q(0) = 0$ and

$$(AF)(a + cu) = (AF)(a) + p^{2s+1}u + p^{2s+2}Q(u).$$

Since A is integral, $(AF)(a) \in p^r R^d$. The strict integer inequality $r > 2s$ permits $k = r - (2s + 1) \geq 0$ and a vector $C \in p^k R^d$ such that $(AF)(a) = p^{2s+1}C$. The normalized root equation is therefore

$$u = -C - pQ(u). \tag{4.13}$$

Integral polynomial evaluation is 1-Lipschitz for the supremum p-adic metric: a common divisibility of input differences is preserved under addition and multiplication by integral elements. Consequently the right side of (4.13) has contraction constant $|p|_p < 1$ on the complete space R^d. It has a unique fixed point u.

To obtain the sharper displacement, begin with $u \in R^d$. If $u \in p^n R^d$, then $Q(0) = 0$ and polynomial congruence give $Q(u) \in p^n R^d$. For $n < k$, both C and $pQ(u)$ are divisible by p^{n+1}. Equation (4.13) therefore improves the divisibility of u by one digit. Induction gives $u \in p^k R^d$. Thus $b = a + p^{s+1}u$ satisfies (4.12), because $s + 1 + k = r - s$. Its normalized preconditioned value vanishes, and injectivity of A gives $F(b) = 0$.

Every competing root $z \in a + p^{s+1}R^d$ can be written $z = a + p^{s+1}w$. Its root equation implies the same fixed-point equation after cancellation of the nonzero scalar p^{2s+1}. Contraction uniqueness gives $w = u$, proving uniqueness on the entire stated ball. □

The existence and uniqueness statements are `Komorebi.PolynomialWeightedLift.exists_weighted_root` and `Komorebi.PolynomialWeightedLift.exists_unique_weighted_root`. The inverse hypothesis in Lean specifies an integral matrix whose image is exactly $p^s J^{-1}$; it does not substitute a potentially larger determinant valuation for s.

Example 4.17 (Strict threshold and displacement boundary)**.** For $p = 2$, $F(X) = X^2 + 3$, and $a = 1$, one has $F(a) = 4$ and $DF(a) = 2$. Thus $s = 1$ and $r = 2$ satisfy every displayed hypothesis except the strict inequality. No root exists in $\mathbb{Z}_2$, since none exists modulo eight. Hence $r \geq 2s$ cannot replace $r > 2s$ uniformly. Also, $F(X) = p^s X + p^r$ has root $-p^{r-s}$, whose displacement is not divisible by p^{r-s+1}. The guaranteed displacement cannot uniformly gain a digit.

4.8 First-order coefficient and parameter motion

Theorem 4.18 (Weighted coefficient deformation)**.** *Let $F, H \in R[X_1, \ldots, X_d]^d$, $F(a) = 0$, and $J = DF(a)$. Suppose $J \in \mathrm{GL}_d(\mathbb{Q}_p)$, $p^s J^{-1} \in M_d(R)$, and $M > 2s$. For $G = F + p^M H$, there is exactly one root $b \in a + p^{s+1} R^d$. It satisfies $b - a \in p^{M-s} R^d$ and, in $\mathbb{Q}_p^d$,*

$$b - a = -p^M J^{-1} H(a) + \varepsilon, \qquad \varepsilon \in p^{2M-3s} R^d. \tag{4.14}$$

Only the original Jacobian $DF(a)$ is subject to an inverse hypothesis.

Proof. Retain the original certificate $A = p^s J^{-1}$ and put $c = p^{s+1}$. Taylor expansion and a polynomial difference identity yield polynomial vectors Q_F, Q_H, both zero at the origin, such that

$$(AF)(a + cu) = p^{2s+1} u + p^{2s+2} Q_F(u), \qquad (AH)(a + cu) = AH(a) + c Q_H(u).$$

Consequently

$$(AG)(a + cu) = p^{2s+1}\big(C + u + pQ(u)\big),$$

where

$$C = p^{M-(2s+1)} AH(a), \qquad Q = Q_F + p^{M-s-1} Q_H, \qquad Q(0) = 0.$$

All exponents are nonnegative. The same contraction and divisibility argument as in Theorem 4.16 gives the root, the displacement p^{M-s}, and full-ball uniqueness. This constructs the root without introducing an inverse hypothesis for $DG(a)$.

For the error estimate, let $\delta = b - a$ and $D = M - s$. Taylor's theorem for the preconditioned polynomial gives

$$E_F = (AF)(b) - (AF)(a) - p^s \delta \in p^{2D} R^d.$$

Polynomial congruence gives $E_H = (AH)(b) - (AH)(a) \in p^D R^d$. The equations $F(a) = 0$ and $G(b) = 0$ imply

$$p^s \delta + p^M AH(a) = -E_F - p^M E_H \in p^{2D} R^d,$$

because $M + D \geq 2D$. Cancel p^s in the integral domain R to obtain

$$\delta + p^{M-s} AH(a) \in p^{2M-3s} R^d.$$

This integral vector is ε. After inclusion into $\mathbb{Q}_p$, $p^{M-s} A = p^M J^{-1}$ gives (4.14). □

The complete theorem is `Komorebi.PolynomialRootMotion.exists_unique_root_motion`. The error depth exceeds the displacement depth by $M - 2s \geq 1$. It is a guaranteed divisibility, not an assertion of exact valuation for every system.

Theorem 4.19 (Simultaneous coefficient and parameter deformation)**.** *Let $F, H \in R[Y_1, \dots, Y_e, T_1, \dots, T_k]^e$ and $a = (y_0, t_0) \in R^{e+k}$ satisfy $F(a) = 0$. Define the actual blocks*

$$J = D_Y F(a) \in M_e(R), \qquad K = D_T F(a) \in M_{e\times k}(R).$$

Suppose $J \in \mathrm{GL}_e(\mathbb{Q}_p)$, $p^s J^{-1} \in M_e(R)$, and $M > 2s$. For every $v \in R^k$, set

$$\tau = t_0 + p^M v, \qquad G = F + p^M H.$$

There is exactly one $y \in y_0 + p^{s+1}R^e$ with $G(y, \tau) = 0$. It satisfies

$$\begin{aligned} &y - y_0 \in p^{M-s}R^e, \\ &y - y_0 = -p^M J^{-1}\big(H(a) + Kv\big) + \varepsilon, \qquad \varepsilon \in p^{2M-3s}R^e. \end{aligned} \tag{4.15}$$

The equality involving J^{-1} is in $\mathbb{Q}_p^e$, with an integral error vector. The parameter equality is exact.

Proof. Use the augmented polynomial systems

$$\mathcal{F}(Y, T) = (F(Y, T), T - t_0), \qquad \mathcal{H}(Y, T) = (H(Y, T), -v).$$

Then $\mathcal{F}(a) = 0$, and its derivative is

$$B = D\mathcal{F}(a) = \begin{pmatrix} J & K \\ 0 & I_k \end{pmatrix}.$$

For $A = p^s J^{-1} \in M_e(R)$ and $c = p^s$, define

$$C = \begin{pmatrix} A & -AK \\ 0 & cI_k \end{pmatrix}.$$

The identities $AJ = JA = cI_e$ give $CB = BC = cI_{e+k}$. In the second product, the upper right block is $-JAK + cK = 0$; in the first, it is $AK - AK = 0$. Thus augmentation preserves the original inverse bound s.

Theorem 4.18 gives a unique root b of $\mathcal{F} + p^M\mathcal{H}$ in $a + p^{s+1}R^{e+k}$, with displacement depth $M - s$. Its last equations say exactly $b_T = t_0 + p^M v = \tau$. Its first equations are $G(b) = 0$. The integral first-order expression uses

$$C\begin{pmatrix} H(a) \\ -v \end{pmatrix} = \begin{pmatrix} A(H(a) + Kv) \\ -p^s v \end{pmatrix}.$$

Projection onto the solved coordinates gives

$$b_Y - y_0 + p^{M-s}A(H(a) + Kv) \in p^{2M-3s}R^e.$$

The same inclusion identity as before converts this to (4.15).

Finally, if $z \in y_0 + p^{s+1}R^e$ is any competing solved root with parameters τ, then (z, τ) belongs to the full augmented ball. Indeed, $M > 2s$ implies $M \geq s + 1$. Full-point uniqueness therefore gives $(z, \tau) = b$. This comparison proves uniqueness on exactly the solved ball stated in the theorem. □

The exact solved-coordinate statement is `Komorebi.PolynomialJointDeformation.exists_joint_deformation`, with its unique-existence interface `Komorebi.PolynomialJointDeformation.exists_unique_joint_deformation`. The triangular inverse identities hold over arbitrary commutative rings, so they apply in particular over the p-adic integers.

Example 4.20 (First-order sign and error precision)**.** For $p = 2$, $s = 0$, $M = 2$, $F(Y, T) = Y + T$, $H = 1$, $a = 0$, and $v = 1$, the prescribed parameter is $\tau = 4$ and the solved root is $y = -8$. Here $J = K = 1$, and the forcing term is $H(a) + Kv = 2$. Omitting the parameter term would predict -4, and reversing its sign would predict 0. The corresponding errors -4 and -8 are both outside $2^{2M}\mathbb{Z}_2 = 16\mathbb{Z}_2$. Thus either change would violate the stated error bound.

The error exponent cannot uniformly be increased. For $F(Y, T) = Y + Y^2 + T$, $H = 0$, $a = 0$, $p = 2$, $s = 0$, $M = 1$, and $v = 1$, there is an even root y of $y + y^2 + 2 = 0$. Its equation forces $y \equiv 2 \pmod 4$ and $y + 2 = -y^2$, so

$$4 \mid y + 2, \qquad 8 \nmid y + 2.$$

Thus the error depth $2M - 3s = 2$ is attained even for parameter-only motion. The same quadratic with a constant coefficient perturbation gives the coefficient-only boundary. These examples refute a uniform extra error digit; they do not assert optimality for every individual choice of p, s, M.

The combined results establish finite singular obstructions, smooth compatible towers, full continuous root charts after a suitable coordinate choice, and quantitative deformation for a rationally invertible solved Jacobian. Rationally rank-deficient persistence remains a separate problem. The analytic solution-map theorem in the next section supplies the regularity conclusion for these same derivative blocks.

5 Analytic regularity of the local solution map

The deformation estimates control how far an exact root moves under a small perturbation. We now show that its local dependence on the parameters is analytic, and compute the derivative from the polynomial Jacobian. The analytic argument takes place over a complete normed field. For integral p-adic data, we first include the coefficients and base point in $\mathbb{Q}_p$; $\mathbb{Z}_p$ itself is not a vector space over $\mathbb{Q}_p$.

Let $\mathbb{K}$ be a complete nontrivially normed field. Let $e, k \in \mathbb{N}$, and write the variables as $(Y, T) \in \mathbb{K}^e \times \mathbb{K}^k$. For

$$F \in \mathbb{K}[Y_1, \dots, Y_e, T_1, \dots, T_k]^e, \qquad a = (y_0, t_0),$$

define the matrices of evaluated formal partial derivatives

$$J = D_Y F(a) \in M_e(\mathbb{K}), \qquad K = D_T F(a) \in M_{e\times k}(\mathbb{K}). \tag{5.1}$$

Rows index equations and columns index variables. We first establish that these formal derivatives are the derivatives of the evaluation map.

Lemma 5.1 (Derivative of polynomial evaluation)**.** *For $P \in \mathbb{K}[X_1, \dots, X_n]$, polynomial evaluation is analytic at every point. At $a \in \mathbb{K}^n$ its strict Fréchet derivative is*

$$v \longmapsto \sum_{j=1}^{n} (\partial_j P)(a) v_j.$$

For a finite system, its strict derivative is the continuous linear map represented by the matrix of evaluated formal partial derivatives.

Proof. For the derivative statement, use polynomial induction on constants, addition, and multiplication by a variable. Constants have zero derivative. The sum rule agrees with the additivity of formal partial differentiation. For PX_j, the product rule gives

$$d(PX_j)_a(v) = a_j\, dP_a(v) + P(a)v_j,$$

which agrees with the evaluated formal product rule for every coordinate. The finite product rule for maps to coordinate spaces then assembles the system derivative. Each matrix defines a continuous linear map because the coordinate spaces are finite-dimensional over the complete field. Polynomial analyticity follows from the corresponding polynomial induction in Mathlib and its finite-coordinate assembly. □

The derivative bridge and analytic statements are `Komorebi.PolynomialAnalyticEvaluation.hasStrictFDerivAt_evalSystem`, `Komorebi.PolynomialAnalyticEvaluation.analyticAt_evalSystem`, and `Komorebi.PolynomialAnalyticEvaluation.analyticOnNhd_evalSystem`.

Lemma 5.2 (The augmented derivative)**.** *If $J \in \mathrm{GL}_e(\mathbb{K})$, the polynomial map*

$$\Phi(Y,T) = (F(Y,T),T)$$

has invertible strict derivative at a, with

$$D\Phi(a) = \begin{pmatrix} J & K \\ 0 & I_k \end{pmatrix}, \qquad D\Phi(a)^{-1} = \begin{pmatrix} J^{-1} & -J^{-1}K \\ 0 & I_k \end{pmatrix}. \tag{5.2}$$

The local inverse supplied by this derivative is analytic near $\Phi(a)$.

Proof. The lower coordinates of Φ are the parameter variables. Differentiating them gives the blocks 0 and I_k. Lemma 5.1 identifies the remaining blocks with (5.1). Multiplication of the two matrices in (5.2), in both orders, gives the identity. The resulting linear equivalence is continuous, as is its inverse. Apply the strict inverse-function theorem to Φ. The analytic inverse theorem applies to the same open partial homeomorphism, because Φ is analytic and its derivative is that continuous linear equivalence. It supplies a convergent local power-series representation of this inverse. □

The augmented Jacobian is identified in `Komorebi.PolynomialJointSystem.jacobian_augmented`. The continuous inverse block and its restriction are `Komorebi.PolynomialAnalyticBlocks.upperEquiv` and `Komorebi.PolynomialAnalyticBlocks.inverse_solved_zero`. Analyticity of the same constructed inverse is `Komorebi.PolynomialAnalyticInverse.analyticAt_inverse`.

Theorem 5.3 (Analytic local root graph)**.** *Suppose $F(a) = 0$ and $J \in \mathrm{GL}_e(\mathbb{K})$. There are open neighborhoods $t_0 \in U \subseteq \mathbb{K}^k$ and $y_0 \in V \subseteq \mathbb{K}^e$, and a map $\psi : \mathbb{K}^k \to \mathbb{K}^e$, such that*

$$\psi(t_0) = y_0, \qquad \psi(U) \subseteq V, \qquad \psi \text{ is analytic at every point of } U, \tag{5.3}$$

$$F(y,t) = 0 \iff y = \psi(t) \qquad (t \in U,\ y \in V). \tag{5.4}$$

At the base parameter, the strict Fréchet derivative is

$$\boxed{D\psi(t_0) = -J^{-1}K.} \tag{5.5}$$

In particular, ψ is differentiable on U.

Proof. Let g be the analytic local inverse of Φ supplied by Lemma 5.2. Since $F(a) = 0$, $\Phi(a) = (0, t_0)$ and $g(0, t_0) = a$. Define

$$\psi(t) = \pi_Y g(0, t),$$

where π_Y projects to the solved coordinates. This gives the base value and analyticity at t_0. The chain rule and (5.2) give

$$D\psi(t_0)v = \pi_Y \begin{pmatrix} J^{-1} & -J^{-1}K \\ 0 & I_k \end{pmatrix} \begin{pmatrix} 0 \\ v \end{pmatrix} = -J^{-1}Kv.$$

To obtain the root equivalence on a product neighborhood, we choose the neighborhoods as follows. The source of the open partial homeomorphism defined by Φ is open around a; its inverse is g. Choose open sets V and U_0 containing y_0 and t_0 such that $V \times U_0$ lies in that source. Restrict U_0 further by the preimage of an open analytic inverse neighborhood under $t \mapsto (0, t)$, the open locus where this same ψ is analytic, and the interior of $\psi^{-1}(V)$. All of these sets contain t_0. For the last set, use the continuity of ψ at t_0 and $\psi(t_0) = y_0 \in V$. Their intersection is an open neighborhood U with $\psi(U) \subseteq V$.

If $t \in U$, $y \in V$ and $F(y, t) = 0$, then $\Phi(y, t) = (0, t)$. The left inverse identity forces $g(0, t) = (y, t)$, hence $y = \psi(t)$. Conversely, the right inverse identity $\Phi(g(0, t)) = (0, t)$ says both that its solved equations vanish and that its parameter coordinates are exactly t. Thus $g(0, t) = (\psi(t), t)$ and $F(\psi(t), t) = 0$. Analyticity at every point of U implies differentiability there. □

This is `Komorebi.PolynomialAnalyticImplicit.exists_analytic_implicit`. Differentiability is also included explicitly in the statement contract `Komorebi.PolynomialRegularityContracts.actual_padic_analytic_root_graph`.

Theorem 5.4 (Regularity of a supplied root selection)**.** *Under the hypotheses of Theorem 5.3, let $h : \mathbb{K}^k \to \mathbb{K}^e$ be continuous at t_0, with $h(t_0) = y_0$. Suppose $F(h(t), t) = 0$ for every t in some neighborhood of t_0. Then h and ψ agree in a neighborhood of t_0. Consequently h is analytic there as a germ and has strict derivative $Dh(t_0) = -J^{-1}K$.*

Proof. Continuity at t_0 places $h(t)$ inside V for all sufficiently close parameters. Intersect this neighborhood with U and the neighborhood on which the root equation holds. Equation (5.4) gives $h(t) = \psi(t)$ throughout the intersection. Analyticity and strict differentiability at a point are unchanged by agreement on a neighborhood. □

The formal result is `Komorebi.PolynomialAnalyticSelection.continuous_selection_regularity`. Its formulation derives $F(a) = 0$ from the root equation near the base parameter and the specified base value.

Corollary 5.5 (Regularity throughout an open parameter set)**.** *Let $U \subseteq \mathbb{K}^k$ be open, and let h be continuous on U. Suppose that for every $t \in U$,*

$$F(h(t), t) = 0, \qquad J_t = D_Y F(h(t), t) \in \mathrm{GL}_e(\mathbb{K}).$$

Then h is analytic at every point of U and its strict derivative is

$$Dh(t) = -J_t^{-1} D_T F(h(t), t) \qquad (t \in U).$$

Proof. At each $t \in U$, openness turns continuity on U into continuity at t and makes the root equation hold on a neighborhood of t. Apply Theorem 5.4 at $(h(t), t)$. □

This is `Komorebi.PolynomialAnalyticSelection.continuous_selection_analyticOnNhd`. The pointwise invertibility assumption is part of this corollary.

Corollary 5.6 (Integral coefficients and centers)**.** *Let p be any prime. Suppose $F \in \mathbb{Z}_p[Y,T]^e$, $a \in \mathbb{Z}_p^{e+k}$, and $F(a) = 0$. Let $\iota : \mathbb{Z}_p \hookrightarrow \mathbb{Q}_p$ be the inclusion. If $\iota(D_Y F(a)) \in \mathrm{GL}_e(\mathbb{Q}_p)$, the polynomial system obtained by including the original coefficients in $\mathbb{Q}_p$ has the analytic local root graph of Theorem 5.3 at $\iota(a)$. The strict derivative at the base is*

$$-\iota(D_Y F(a))^{-1}\iota(D_T F(a)).$$

The open neighborhoods and the output map are in the ambient spaces over $\mathbb{Q}_p$.

Proof. Coefficient inclusion commutes with evaluation and formal partial differentiation. Consequently both Jacobian blocks after inclusion are the entrywise inclusions of the original blocks, and the original exact root remains an exact root. Apply Theorem 5.3 to the system with included coefficients. □

This is `Komorebi.PolynomialRegularityIntegral.exists_analytic_integral_implicit`. It requires no invertibility modulo p. In particular, it includes solved Jacobians singular modulo p but invertible over the fraction field.

Example 5.7 (The derivative sign and a nonlinear control)**.** For $F(Y,T) = Y + T$, the exact branch is $\psi(T) = -T$ and $J = K = 1$. The derivative is -1. Uniqueness of the derivative rules out $+1$ over $\mathbb{Q}_p$, including $p = 2$. The nonlinear polynomial $F(Y,T) = Y + Y^2 + T$ has $F(0,0) = 0$ and $J = K = 1$ there. Applying the formal theorem to this polynomial constructs an analytic local root graph with derivative -1. Thus the example checks the existence conclusion as well as the derivative formula. The declarations and the deliberately rejected positive-sign statement are in the regularity control and statement-contract files.

The results cover arbitrary finite e and k, including zero. The local inverse machinery explicitly handles a zero-dimensional domain. The analytic theorem does not assign a uniform numerical radius to its neighborhoods. The earlier weighted lifting and joint deformation theorems retain their quantitative radii and error bounds, while the present statements establish regularity of the local solution germ. A direct Lean identification with each historical subtype-valued chart is an additional interface task. General persistence at a fraction-field singular solved block is a separate problem.

6 Open questions and further directions

The arithmetic results above resolve Nishibuchi's conjecture throughout its stated parameter range. The local analyticity and differentiability theorem is also complete under its stated invertibility hypothesis. What remains is a set of extensions and sharper classifications, together with the broader questions raised by the original native seed-calculus program.

6.1 Singular persistence beyond an invertible solved block

The weighted lifting theorem allows the solved Jacobian to be noninvertible modulo p provided that it is invertible over $\mathbb{Q}_p$ with a controlled inverse. This includes a substantial class of lifts often called singular modulo p. The next problem is to understand Jacobians that remain rank-deficient over $\mathbb{Q}_p$, where the inverse construction is no longer available.

For a fixed polynomial system $F \in \mathbb{Z}[X_1, \ldots, X_d]^e$, a precision $L \geq 1$, and a root $\bar{x}$ modulo p^L, let

$$\mathcal{L}_{F,L}(\bar{x}) = \{x' \in (\mathbb{Z}/p^{L+1}\mathbb{Z})^d : F(x') = 0,\ x' \equiv \bar{x} \pmod{p^L}\}.$$

The finite obstruction theorem decides whether this set is empty from the residual digit and reduced Jacobian. When it is nonempty, the theorem also determines its cardinality and affine kernel structure. The conclusion concerns one level: vanishing of the obstruction permits the next step, but does not guarantee that subsequent steps are possible.

For example, $X^2 - 5$ has the root 1 modulo 2, and both 1 and 3 lift it to roots modulo 4. Every odd square is 1 modulo 8, so neither branch continues to a root modulo 8. Thus even a successful first correction can lead to branches that stop at the following level.

We seek structural hypotheses and higher obstruction data that control an entire singular branch, including its behavior under coefficient and parameter deformation. One route is to propagate suitable conditions from level to level. Another is to prove existence at all levels in a form that yields compatible branches. The finite obstruction calculus supplies the tests for individual levels, but a persistence theorem needs a further argument linking them. Any general formulation must allow multiple branches, later obstructions, and changes in the rank of the Jacobian.

6.2 Native 13.2: presentation invariance

The original conjecture asks for a canonical finite obstruction packet whose vanishing is necessary and sufficient for a one-level lift in a finitely presented arithmetic seed system. For a fixed finite polynomial presentation, the paper now supplies the precise obstruction

$$o_{F,\bar{x}} = [r] \in \operatorname{coker}\Big(J_{F,\bar{x}} : \mathbb{F}_p^d \longrightarrow \mathbb{F}_p^e\Big),$$

where r is the residual digit. We have proved independence from the chosen integer representative. Invariance under a change of equations or presentation requires more.

The raw cokernel already reveals the difficulty. The presentations $F = (X)$ and $G = (X, 0)$ describe the same zero locus over every coefficient ring. At zero their Jacobians are, respectively,

$$\mathbb{F}_p \xrightarrow{1} \mathbb{F}_p, \qquad \mathbb{F}_p \xrightarrow{(1,0)} \mathbb{F}_p^2.$$

Their raw cokernels are 0 and $\mathbb{F}_p$. Both distinguished obstruction classes vanish, but the ambient obstruction groups are not isomorphic. The example rules out the raw cokernel as a presentation-independent object while leaving open the possibility of an intrinsic obstruction construction.

A precise form of Native 13.2 therefore needs four ingredients:

1. a category of seed objects, including how the operators, relations, projectors, weights, and topology enter an object;
2. specified changes of presentation and a proof that they preserve the actual lift problem;
3. an obstruction object that respects dependencies among the relations, together with comparison maps or a universal property;
4. a precise meaning of “finite” and “canonical”, and a proof that vanishing is equivalent to the intended one-step continuation.

For commutative algebra presentations, the relation module offers a natural way to account for dependencies among listed equations. To extend this approach to the full seed vocabulary,

we would need an encoding that preserves the lift problem, together with a proof of that preservation. This is one of the links still needed between the concrete obstruction theorem and Native 13.2.

6.3 Native 13.3: finite compatibility, completion, and descent

The original proposed form is

$$X \simeq X_2^{\mathrm{wt}} \times_C {\prod_{p>2}}' X_p.$$

It leaves X, the distinguished subsets defining the restricted product, the maps to C, the ambient category, and the universal meaning of "smallest" unspecified. In particular, one must decide whether X is a global object or a completed object.

The linear compatibility theorem provides one concrete part of this program. If two local observations $J : U \to V$ and $K : U \to W$ are linear maps from the *same* correction module, define

$$C_{J,K} = U/(\ker J + \ker K).$$

The attainable observations $\operatorname{im} J$ and $\operatorname{im} K$ have canonical labels in this quotient. Equality of labels is equivalent to simultaneous realization by a single correction. The joint image is exactly the fiber product over $C_{J,K}$, with the proved linear universal property. The declarations are `Komorebi.SharedCorrection.labels_eq_iff_common`, `Komorebi.SharedCorrection.joint_range_eq_fiber`, and `Komorebi.SharedCorrection.common_universal`.

To apply this result arithmetically, one must construct the common correction module and the observation maps, then show that the resulting compatibility condition agrees with the intended arithmetic one.

Why the global/completed distinction is necessary. Consider constant sequences with integer initial value. Their global parameter space is $\mathbb{Z}$, while their local parameter spaces are $\mathbb{Z}_p$. Under the canonical local maps, the image of the global objects is the diagonal

$$D = \{(n)_p : n \in \mathbb{Z}\} \subseteq \prod_p \mathbb{Z}_p.$$

This diagonal is dense: a basic open set specifies finitely many congruences, and the Chinese remainder theorem provides an integer satisfying them. It is proper: the tuple with 2-adic coordinate zero and all odd-adic coordinates one is not the image of an integer. Indeed an integer zero in $\mathbb{Z}_2$ must be zero, contradicting its 3-adic coordinate.

Consequently this canonical global diagonal cannot be an equality locus

$$\{(a, b) : f(a) = g(b)\} \subseteq \mathbb{Z}_2 \times \prod_{p>2} \mathbb{Z}_p$$

for continuous maps to a Hausdorff space C. Such an equality locus is closed, as the inverse image of the closed diagonal in $C \times C$, whereas D is proper and dense. This topological argument, retained from the native progress note, concerns the stated topological interpretation. It has not been formalized in Lean for this manuscript.

Finite CRT reconstruction leads to the familiar completed object $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, but does not identify which local tuples come from a single global integer. A complete form of Native 13.3 must therefore specify finite-level compatibility, transition maps and completion, and any additional global descent condition. If recurrence, spectral, or Frobenius information is to supply that condition, it must be expressed through defined maps or constraints on the local data, with a precise account of how they enter C.

The accompanying `native/Native_Conjectures_Progress.md` records further results in restricted settings: an intrinsic finite Ext^1 obstruction for finitely presented commutative integer algebras, finite CRT reconstruction for polynomial solution sets, and a characterization of rectangular finite relations by equality of common labels. The obstruction result is a stated specialization of a classical external theorem; the CRT and rectangularity results have prose proofs in the note. These are preserved mathematical progress, with the restrictions and source records given there, rather than Lean completions of the general Native 13.2 and 13.3 statements.

6.4 Arithmetic refinements

Write $U_{N,j}(n) = j!C_{N,j}(n)$ and $U_j = U_{4,j}$, and use $v_p(0) = \infty$ when discussing valuation bounds. The following classification questions remain open in this program.

1. **Higher dyadic differences.** Determine the exact value of
$$v_2\Big(\Delta_u^k U_j(n)\Big), \qquad k \geq 2,$$
with explicit treatment of zero differences. The all-order lower bounds and the exact first-displacement theorem do not determine this value: subtraction of two first differences can introduce further cancellation. The first concrete target is order 2 in one fixed address.

2. **Least periods beyond $N = 4$.** For further packet families, determine the least positive P for which $U_{N,j}(n + P) \equiv U_{N,j}(n) \pmod{p^r}$ for every n in the claimed range. The general sufficient periods are proved. Their minimality does not follow from those sufficient bounds, and a period in packet index is a degree shift of NP.

3. **Exact affine Mahler-kernel valuations.** For the kernel of Section 2.7, determine exact values of
$$v_p(\Theta_{M,s}(r,k)), \qquad \Theta_{M,s}(r,k) = [z^r](1+z)^s((1+z)^M - 1)^k,$$
beyond the explicit special families. The minimum valuation of the summands gives a lower bound. Equality requires a separate argument excluding cancellation.

4. **Exact classical dyadic tail values.** The uniform truncation exponent $4M + 3$ and its optimality are proved. Individual term weights are also known. The valuation of the whole omitted tail at an arbitrary even row may be higher, so an exact formula for all rows requires more than taking the minimum of those term weights. Empty tails must remain exactly zero.

These refinements build on the proved estimates and exact cases, including the sharp uniform classical dyadic bound and the resolution of Nishibuchi's conjecture.

6.5 Historical material and interface work

The original Paper IX, with corrected proof-status labels, survives as a 28-page PDF. Its recovered formalization supports the framework, Euler coordinates, Mahler differences, unit coordinates, recurrence packets, and CRT results retained in this manuscript. The original TeX, original figure-generation files, and complete historical experiment package were not recovered with that PDF. Their older figure and test counts remain historical records; those computations have not been reproduced for this consolidation.

A reconstructed textual copy of the Annals manuscript was also recovered. It includes finite-field mode and subgroup inversion, Fourier invariance, Carmichael and matrix transfer,

Amice pullbacks, depleted analytic Euler branches, and extensive polyphase formulas. The sharp classical dyadic kernel and truncation claims from that text have since been connected to checked Lean proofs. The finite polyphase formulas used for explanation here have short explicit algebraic derivations. The remaining claims in that larger historical text have not all been formalized. The source archive retains the text with its provenance, and the statement map distinguishes these historical claims from the certified claims in the present manuscript.

Remaining recovery work includes the precise older odd-prime edge identities, their stated ranges, and the binary digit-sum form of the checked weights. For each item, the work is to recover the formula with its stated range, check the proof, and identify any corresponding Lean declaration. Where the historical source is still missing, the record remains incomplete.

One further task is to connect the older chart constructions to the new regularity theorem. That theorem applies to polynomials with arbitrary coefficients in the ambient field $\mathbb{Q}_p$ and identifies the analytic germ of every continuous ambient-field root selection under its hypotheses. Several older rectangular charts were defined using integer coefficients, integer centers, or subtype-valued maps. Comparing each such historical interface explicitly with the new ambient formulation remains to be done. These comparisons would connect the existing constructions to the proved analytic implicit-function theorem. General singular persistence remains the separate problem described in Section 6.1.

7 Formal evidence and reproduction

The verification record supports the manuscript in several ways. The Lean source states the quantified theorem and its hypotheses. A compiler report records acceptance of that source by a specified Lean and Mathlib environment. An axiom audit identifies the axioms used by the resulting named declarations. Statement contracts and mathematical controls check selected interfaces and boundary cases. Finally, source correspondence reviews compare the formal objects with the intended mathematical statement. Each check has a specific role: numerical examples test particular cases, whereas a universal claim requires a proof, and correspondence with an external conjecture requires a comparison of the statements.

This edition revises the exposition and checks the evidence index against the preserved sources. The Lean sources are unchanged. Their compiler acceptance is supported by the earlier successful reports, whose source and log hashes are checked by the accompanying index. The most recent compilation covers local analytic regularity and its polynomial dependencies; earlier reports cover the arithmetic and native obstruction developments.

The proofreading pass also restores several results omitted from the first consolidation and records the comparison with the original paper. Its coverage matrix and semantic review appear in `consolidation/editorial/`. The Bernoulli boundary is accompanied by a classical derivation, without a claim of a dedicated Lean certificate. The reviews distinguish this addition from restored results already present in the checked sources.

7.1 Verification frontiers

Table 1 gives the recorded scope of each major verification frontier. An *inventoried module* is a source file in that frontier's formalization list. Older lists sometimes include files of examples or positive controls in this group. A *check* is an entry in the report's run list, including expected rejection of invalid claims. An *axiom closure* is the recorded axiom dependency set of an explicitly named declaration; definitions and abbreviations contribute to this count as well as theorems. None of these columns is a count of distinct mathematical results across the paper.

The baseline inventory contains 836 theorem commands, 162 definitions and two abbrevia-

Preserved frontier	Modules	Checks	Axiom closures
Original completion baseline	79	$79 + 37 + 3$	1000
Dyadic strengthening and isometry	22	38	246
Standalone finite affine obstruction	2	4	17
Shared correction quotient	6	13	60
Polynomial one-step lifting	9	18	91
Exact polynomial lift counts	9	17	106
Smooth infinite towers	14	22	127
Coefficient deformation	17	26	159
Continuous parameter charts	25	40	244
Weighted lifting	30	50	304
First-order root motion	35	58	326
Prescribed parameter obstruction	39	65	375
Prescribed parameter towers	45	74	408
Joint coefficient and parameter motion	49	82	460
Local analytic regularity	55	91	515

Table 1: Recorded verification scopes. Rows overlap extensively through inherited sources and must not be added. The baseline separates 79 module checks, 37 control checks, and three additional recorded runs; its nine dependency revision comparisons are recorded separately.

tions, matching its 1000 audited names. The latest polynomial inventory contains 322 theorem commands, 76 definitions and seven abbreviations in its 55 formalization modules. Its eight positive-control support modules and 13 statement-contract modules bring the full audit to 415 theorem commands, 92 definitions and eight abbreviations, hence 515 closures. The 91 checks also include 12 negative controls. Six new regularity modules contribute 30 theorem commands and eight definitions; all 49 inherited formalization modules are unchanged in that release.

The arithmetic strengthening inventory exposes a small historical metadata error. Its summary fields say 179 theorems and 12 definitions, while its module entries list 215 theorem commands, ten lemma commands, 20 definitions and one abbreviation. These 246 named entries exactly match the preserved axiom audit. The consolidation uses the entries and audit, and records the stale summary fields explicitly. The original inventory remains unchanged as historical evidence. This is a correction to the reporting metadata, not a newly discovered proof or a modification of a checked theorem.

7.2 Claim-to-source correspondence

The source references attached to the mathematical sections identify the principal declarations for each result. Their complete names, exact source hashes, roles and verification frontiers are collected in `consolidation/audit/DECLARATIONS.csv`. The accompanying `consolidation/audit/EVIDENCE_INDEX.json` records the corresponding inventories, reports and retained logs. In particular, the following interfaces distinguish the principal completed results from generic auxiliary lemmas:

- The external conjecture is assembled by `NishibuchiCompletion.conjecture_3_9` in the original completion baseline. Its coefficient identification, integrality, parameter split and explicit threshold are part of that proof chain.
- Exact $N = 4$ packet periods are supplied by `DyadicFourLeastPeriods.least_packet_period`; the isometric extension is supplied by `DyadicFourIsometry.existsUnique_isometry`. The normalization of the original coefficients is an explicit part of these interfaces.

- The classical sharp edge tower is supplied by `DyadicSharpTower.dyadic_supercongruence_sharp`, together with its first omitted error valuation. Its optimality concerns the stated uniform truncation bound.
- Prescribed parameter lifting uses the actual residual and actual partial Jacobians in `PolynomialPrescribedObstruction.prescribed_lift_iff_obstruction_zero`. The quotient criterion and exact lift counts are not assumptions of the later tower results.
- Joint deformation is supplied by `PolynomialJointDeformation.exists_unique_joint_deformation`. Local regularity is supplied by `PolynomialAnalyticImplicit.exists_analytic_implicit`, with the actual polynomial derivative and the derivative formula $-J^{-1}K$.
- `PolynomialAnalyticSelection.continuous_selection_regularity` transfers the analytic germ and strict derivative to any continuous local root selection satisfying the stated root and base-point conditions.

Here arithmetic declaration names have namespace prefix `KomorebiPaperIX.`; polynomial and native declarations have prefix `Komorebi.`. The section maps and CSV retain the full names.

The declaration index covers every explicitly named theorem, lemma, definition and abbreviation in the 14 release-frontier inventories, together with the earlier standalone affine report. Byte-identical copies of a source file are collapsed, while all their locations remain recorded. If the same qualified name occurs in different source versions, the versions remain separate and are marked. Anonymous examples, compiler-generated declarations, failed development attempts, and the contents of earlier nested historical snapshots are not counted as named results in this index. Negative-control source files are checked for integrity but are not recorded as proved mathematical assertions. Consequently, the number of CSV rows must not be quoted as a global unique theorem count.

7.3 External statement and library dependencies

The original conjecture correspondence was reviewed against Nishibuchi's arXiv version `2605.10677v2`, dated 1 June 2026. The retained source record identifies the official version, the full title, the PDF checksum and the extracted-text checksum. The detailed review compares Definition 2.1, Notation 3.2 and Conjecture 3.9 with the actual reciprocal EGF coefficients, the factor $j!$, the residue range, the degree displacement, and the threshold in the formal capstone. Nishibuchi's conjecture is the conclusion to be proved; it is not an axiom imported into Lean.

The full external paper is not redistributed in this source release. Consolidation preserves the earlier source correspondence audit and its version record. A fresh request to the official arXiv endpoints was blocked by the retrieval tool, so this edition does not claim a new inspection of the primary paper. This access failure affects fresh reauthentication of the external wording, not the recorded compiler acceptance of the unchanged formal sources. No broad literature or priority claim is made.

The mathematical environment is Lean `4.34.0-rc2`, with Mathlib revision `aeb72f1e9981b6d35b426b16dca7759be3e7ba06`. The allowed axiom dependencies are `propext`, `Classical.choice` and `Quot.sound`. The reports do not describe a rebuild of Mathlib from source. The local regularity audit separately records the applicable Mathlib analytic inverse and strict derivative interfaces and their hypotheses. In particular, the field-valued analysis is performed over a complete nontrivially normed field; its integral specialization embeds the data into $\mathbb{Q}_p$. It does not use $\mathbb{Z}_p$ as a vector space over $\mathbb{Q}_p$.

7.4 Reproduction and limits of the consolidation check

The archive contains the mathematical sources, verifier scripts, controls, contracts, prior logs and exact inventories. The pinned Lean and Mathlib kit is supplied separately. From the consolidated archive root, the latest polynomial gate can be reproduced with a new output directory:

```
python3 replay_paper9_regularity.py \
  --kit /absolute/path/to/pinned-kit \
  --output /absolute/path/to/new-polynomial-run
```

The preserved arithmetic, native and shared-correction chain has a separate entry point:

```
python3 replay_shared_correction.py --all \
  --kit /absolute/path/to/pinned-kit \
  --output /absolute/path/to/new-arithmetic-run
```

These commands reproduce the recorded checks; they were not rerun during manuscript consolidation or proofreading. The second command rebuilds the preserved baseline before the arithmetic continuation. Each output path must be new. The reports retain the original full commands, runtime fingerprints and output hashes. The executable-path compatibility adapter used in the recorded environment restores runtime path discovery; it does not alter Lean's mathematical kernel or the source statements.

For the narrower evidence reconciliation carried out for this edition, use

```
python3 consolidation/tools/evidence_index.py \
  --source . --output /absolute/path/to/new-evidence-index
```

This check compared 644 source-file occurrences with their recorded hashes and 713 retained log occurrences with their recorded hashes. Four additional logs from the older standalone affine report were fingerprinted, but that legacy report does not contain prior log hashes; those four are not promoted to hash comparisons. Every named declaration in the selected inventories matches the appropriate recorded standard-axiom dependency set. These are integrity and provenance checks, not another Lean compilation.

Expected negative-control failures are retained as evidence that specific invalid strengthenings are rejected. Examples include an unjustified extra precision digit, omitted parameter terms, a wrong derivative sign, and uniqueness without the appropriate hypotheses. Development failures and interrupted runs remain historical records; only the successful designated release reports support the verification table. Only accepted declarations in the designated successful reports support claims of formal verification.

Finally, the proof of the original stated conjecture does not resolve every extension mentioned in this manuscript. The open questions concerning rationally rank-deficient singular points, presentation invariance, global or completed descent, and further exact arithmetic have no certificate supplied by these counts. Section 6 states the remaining implications explicitly.

References

[1] Yuta Nishibuchi, *p-adic Congruences of Generalized Euler Numbers and Relations to Even Zeta Values*, arXiv:2605.10677v2, 1 June 2026. https://arxiv.org/abs/2605.10677v2. The comparison to Conjecture 3.9 uses the source review and fixed-version record preserved with this edition.

[2] The Mathlib Community, *Mathlib*, source revision `aeb72f1e9981b6d35b426b16dca7759be3e7ba06`, used with Lean 4.34.0-rc2. In particular, the polynomial, strict inverse-function and analytic inverse-function developments cited in Section 5. The exact local source paths and dependency records accompany the release.

[3] Komorebi P., *Projector-Resolved Seed Calculus IX: original manuscript, formal supplements and cumulative continuation sources*, preserved source collection, consolidated 8 September 2026. The original 28-page status-corrected manuscript and all subsequent release records remain in the accompanying archive.